RUMBA: Russian User Memory Benchmark

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Abstract

The ability to handle long-term memory in LLMs is becoming increasingly critical, yet existing benchmarks remain English-centric and rely on aggregate retrieval metrics, failing to capture interactions between long-range context, temporal information, and reasoning. To address this, we introduce **RUMBA** (Russian User Memory BenchmArk) — a new benchmark for long-term conversational memory that provides a fine-grained taxonomy of memory-centric question types and a unified methodology accounting for semantic type, session scope, temporal reasoning, and the explicitness of temporal expressions. RUMBA consists of timestamped user–assistant dialogues with QA pairs requiring retrieval, combination, and reasoning across sessions. While designed for Russian, we also provide an aligned English subset under the same methodology. We evaluate contemporary memory systems and long-context models, and show how RUMBA serves as a diagnostic tool to analyze model behavior across benchmark slices and identify strengths and failure modes of different memory mechanisms.

1 Introduction

Large language models (LLMs) are increasingly deployed as conversational assistants that must retain user-specific information across sessions and remain consistent over time. This requires more than simple retrieval: models must preserve salient details, resolve long-range references, and reason over temporal updates and multi-step dependencies.

Recent English-centric memory benchmarks (Maharana et al., 2024; Wu et al., 2025; Bei et al., 2026) have advanced long-term memory evaluation. However, recent work on memory-agent evaluation notes that existing memory benchmarks are still largely recall- or retrieval-oriented and provide incomplete coverage of memory behaviours required in realistic long-horizon interactions, including memory updates and consolidation, stale or obsolete information, and forgetting (Du, 2026; Hu et al., 2026; Uddin et al., 2026). As summarizes in Table 1, existing benchmarks also rely on synthetic dialogue construction and offer limited diagnostic granularity through relatively coarse question categories. This makes it difficult to isolate failure factors such as temporal dependencies, conversational span, or query semantics.

Meanwhile, Russian benchmarks such as MERA (Fenogenova et al., 2024) and LIBRA (Churin et al., 2025) evaluate broader model capabilities, including general language understanding and long-context processing, but not conversational memory in a targeted way. This leaves a clear need for a benchmark focused on memory, temporal awareness, and reasoning in long Russian dialogues.

To address this gap, we introduce **RUMBA** (**R**ussian **U**ser **M**emory **B**enchm**a**rk), a benchmark for evaluating long-term conversational memory in Russian. RUMBA is built around long-form dialogue scenarios and a taxonomy of question types covering contextual recall, entity tracking, temporal reasoning, and multi-step reasoning. We provide two test sets (Russian and an aligned English translation) and conduct evaluations under both RAG and full-context scenarios.

Our contributions are as follows:

- We create and release **RUMBA** as an open-source benchmark[1], the first benchmark specifically designed to evaluate long-term conversational memory in Russian, accompanied by an aligned English translation to

[1]https://huggingface.co/datasets/ai-forever/RUMBA

enable cross-lingual diagnostics and reproducibility.

- We present a fine-grained taxonomy of memory-centric question types along multiple axes, enabling detailed model comparison beyond aggregate accuracy and simple retrieval.
- We implement a unified evaluation pipeline that supports both retrieval-based memory systems and full-context baselines for both languages, which standardizes the evaluation workflow while allowing each method family to operate under its natural memory-access setting[2].
- We conduct an extensive empirical analysis of different families of memory solutions and a single-model case study, demonstrating RUMBA as a diagnostic tool for identifying strengths and failure modes across benchmark slices.

We hope that RUMBA will serve both as a diagnostic benchmark for evaluating memory in Russian-language conversational systems and as a practical resource for developing models that can reliably store, update, and use user-specific information over time.

2 Related work

Recent work on evaluating memory capabilities in language models has been shaped by benchmarks such as LoCoMo (Maharana et al., 2024), LongMemEval (Wu et al., 2025), Mem-Gallery (Bei et al., 2026). These datasets provide a structured starting point for studying long-term memory in conversational systems. In particular, LoCoMo organizes evaluation around reasoning-oriented question families (e.g., single-hop, multi-hop, temporal, commonsense, adversarial), while LongMemEval defines a set of core memory abilities (e.g., recall, personalization, multi-session reasoning, updates, temporal reasoning, abstention) and operationalizes them through specific question types. Together, these benchmarks establish a useful conceptual and methodological baseline for memory evaluation and, particularly, for the RUMBA taxonomy.

However, they have three key limitations (Table 1 shows the overview): (1) flat taxonomies that entangle reasoning complexity, temporal structure, and memory access within single question categories; (2) treating temporal reasoning and knowledge updates as isolated task types rather than dimensions that can interact; and (3) ignoring memory control behaviors such as intentional forgetting or validity constraints.

Several benchmarks (e.g., TimeBench (Chu et al., 2024), temporal NLI (Vashishtha et al., 2020), and the three-dimensional temporal QA framework (Piryani et al., 2026)) evaluate temporal reasoning in LLMs, covering event ordering, duration, and temporal commonsense. However, they are not designed for long-context settings nor grounded in realistic user–assistant interactions, and thus fail to capture temporally dependent reasoning across extended multi-session conversations.

The gap is even more pronounced for Russian: no dedicated benchmark for persistent long-term memory exists. LIBRA (Churin et al., 2025) evaluates long-context understanding, but does not test structured memory usage over time. MERA (Fenogenova et al., 2024) focuses on general reasoning and knowledge tasks, without isolating memory-specific abilities. The only partially relevant dataset, ruTiE (Chervyakov et al., 2026), includes conversational evaluation, but operates on short dialogues and does not capture persistent, multi-session memory phenomena.

To address all of the above limitations (flat taxonomies, lack of interactive temporal dimensions, absence of forgetting scenarios, inapplicability to long-context dialogues, and the complete gap for Russian), we propose a benchmark for Russian with an extended taxonomy covering forgetting and broader reasoning types, multi-axis annotations for fine-grained diagnostics, and human-written user turns to reduce synthetic interaction bias.

3 RUMBA Methodology

3.1 Overview

RUMBA was designed around a set of diagnostic research questions that directly address the limitations identified in prior work. The goal is not only to produce an aggregate score but to make model evaluation informative about specific memory capabilities and failure modes. Accordingly, dialogues, questions, answers, and evidence configurations were created to support controlled comparisons along the main dimensions of long-term conversational memory. The benchmark supports the following research questions:

[2] https://github.com/ai-forever/RUMBA

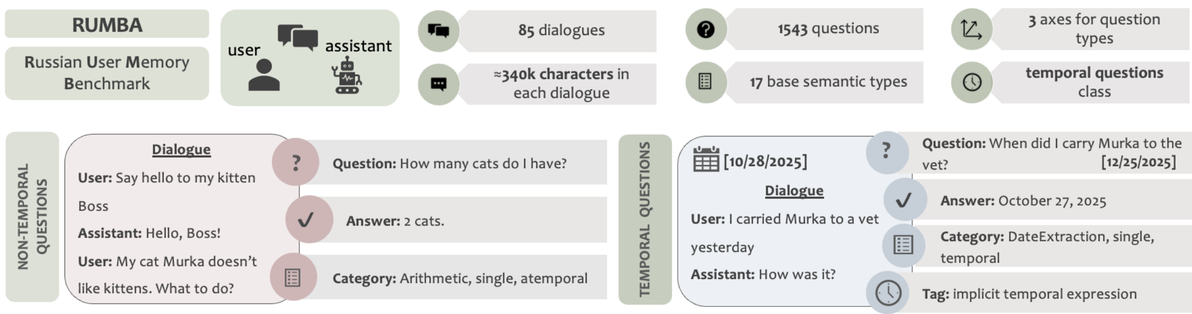

Figure 1: Overview of the proposed benchmark and annotation framework. The figure summarizes dataset-level statistics, representative user–assistant dialogues, example question–answer instances, and the multi-dimensional annotation scheme used to categorize memory questions.

Feature	LoCoMo	LongMemEval	Mem-Gallery	RUMBA
Modality	Text/Image	Text	Text/Image	Text
Language support	English only	English only	English only	**Russian + English**
Context length (chars)	44k–90k	S: 455k–514k / M: 4.5M–5.2M	42k–86k	RU: 297k–569k / EN: 234k–528k
# Annotated Q-A pairs	1,542	500	1,711	1,543
Question taxonomy	Flat (5 types)	Flat (5 types)	Flat (3 types)	**Orthogonal axes** ($17 \times 2 \times 2$ **types**)
User utterances origin	Generated	Generated	Mixed	**Human-written**
Temporal reasoning as independent dimension	No	No	No	**Yes** (✓)
Explicit support for "forgetting"	No	No	No	**Yes** (✓)
Memory scenario coverage	retention, abstention	retention, abstention, updating	retention, abstention, updating	retention, abstention, updating, **forgetting**
Date for each dialogue's utterances	Yes	Yes	Yes	Yes
Date for questions	No	Yes	No	Yes
Failure diagnosis granularity	Limited	Limited	Limited	**Finer-grained (temporal vs. atemporal)**

Table 1: Comparison of memory benchmarks: LoCoMo, LongMemEval, Mem-Gallery, and RUMBA. **Modality**: content types present in the dialogues; **Language support**: languages of the dialogues and questions; **Context length**: dialogue length measured in characters; **# Annotated Q-A pairs**: number of question-answer pairs associated with each dialogue; **Question taxonomy**: structure of the question taxonomy, distinguishing *Flat* taxonomies, where each question is assigned a single type, from *Orthogonal* axes, where each question receives multiple tags; RUMBA uses semantic × quantitative × temporal axes; **User utterances**: the primary source of user-side dialogue utterances, distinguishing LLM-generated, mixed-source, and human-written data; RUMBA uses manually written user utterances. **Temporal reasoning as an independent dimension**: whether one of the assigned tags indicates that a question is temporal or atemporal; **Explicit support for "forgetting"**: whether the benchmark includes scenarios involving deletion requests, such as "delete this" or "forget this"; **Memory scenario coverage**: high-level task categories used for memory evaluation; **Dates for dialogue utterances and questions**: whether each utterance and question is timestamped; **Failure diagnosis granularity**: the ability to identify which benchmark dimension contributes to question complexity.

1. **Memory scope and semantics.** How does performance vary with session scope (single vs. multiple sessions) and with semantic supergroups (extraction, reasoning)?
2. **Temporal reasoning.** How does performance differ between temporal and atemporal questions, and between cases where temporal information is explicit vs. implicit?
3. **Retrieval vs. full-context** How do retrieval-augmented memory systems, which trade off perfect recall for scalability, compare with full-context baselines that preserve all information within a finite window but cannot scale indefinitely?

Fig.1 shows the overall structure of the proposed benchmark.

3.2 Taxonomy

The RUMBA taxonomy is designed to address the limitations discussed in Sec.2 by moving from a flat, benchmark-specific categorization to a multi-dimensional, compositional framework for memory evaluation. Instead of assigning each question to a single class, RUMBA decomposes tasks along several orthogonal axes, including:

- the semantic operation required (e.g., recall, update tracking, comparison),
- the temporal dimension (temporal vs. atemporal),
- the number of sessions involved,
- the reasoning complexity,
- and the memory validity state (e.g., active vs. forgotten information).

The questions are categorized according to three axes: (1) *Semantic axis* represents the classification of questions with respect to their meaning and assessment focus. Broadly, the questions are classified into Information Extraction and Reasoning classes. (2) *Quantitative axis* indicates the number of sessions required to answer a given question. Questions are divided into single-session and multi-session classes. (3) *Axis of temporality* denotes whether a question is temporal. Questions are divided into temporal and atemporal classes. The summary of the axes and corresponding types of questions is presented in Table 2.

This axis-based design enables a more precise characterization of model behavior. For example, the same underlying operation (e.g., recall) can be evaluated in both temporal and atemporal settings, revealing different failure modes. Similarly, reasoning is treated not as a separate category, but as a property that can be combined with different types of memory access, allowing finer-grained analysis of model capabilities.

A key strength of RUMBA is its treatment of *temporality* as an independent dimension. Unlike prior benchmarks, where temporal reasoning is isolated as a specific task type, RUMBA allows temporality to interact with all other axes. This makes it possible to systematically study how models handle time-dependent versus time-independent memory queries within the same semantic framework.

Another important contribution is the explicit modeling of memory control and validity. In addition to standard abilities such as recall and update tracking, RUMBA introduces evaluation of intentional forgetting, where models must correctly exclude previously stored information after explicit user instructions. The forgetting scenario is supported through user behavior and a dedicated question type, DeleteInfo. Users may trigger it with requests such as "delete this" or "forget it." The correct response is "There is no such information" (or similar), indicating that the information has been successfully forgotten and is no longer retained by the system.

This reflects realistic requirements for memory-enabled assistants and introduces a new class of failure modes not captured in prior work. A detailed taxonomy is provided in the Appendix A.

The taxonomy was designed with close consideration of real-world production user behavior in memory-enabled assistant systems. In particular, its structure was informed by observations from a deployed proprietary conversational assistant with long-term memory functionality, including naturally occurring user behaviors related to storing, updating, correcting, and deleting memories. Grounding the taxonomy in authentic interaction patterns allowed us to better capture the diversity and practical characteristics of real conversational memory queries beyond synthetic or narrowly constructed benchmark settings.

Table 2: RUMBA Taxonomy Summary. Example of question's full annotation: $Question$ = What do I like about my job? $Types$ = StaticUser, multi, atemporal. Full taxonomy description is available in Table 26 of Appendix.

Axis	**Categories / Types**
Semantic	Extraction class (6 types): StaticUser, UpdatingInfo, DeleteInfo, AsstQuery, OpenDomainType, DateExtraction Reasoning class (10 types): SocialRelationship, Ordering, Arithmetic, Comparison, UserQA, CalendarUnderstanding, TemporalCommonsense, TemporallyModifiedGeneralReasoning, ComplexRelations, OtherReasoning Abstention type: no answer in dialogue Types with subtypes: OtherReasoning, Arithmetic, DateExtraction, CalendarUnderstanding, TemporalCommonsense, TemporallyModifiedGeneralReasoning, ComplexRelations
Quantitative	Single-session, Multi-session
Temporality	Atemporal, Temporal Temporal Expression Tags: explicit / implicit / no temporal expression

3.3 Dataset characteristics

Table 3 summarized the main characteristics and statistics of the Russian version of the dataset. The instances that comprise the dataset represent textual dialogues between an assistant (LLM) and a user, as well as questions associated with each dialogue that relate to its content.

Sessions. Each dialogue consists of timestamped sessions, each representing one day of interaction with the assistant. We distinguish *evidence sessions*, which contain information relevant to at least one question, from *filler sessions*, which include both conversations about the user that are irrelevant to questions and sessions with no user-related factual information. Evidence sessions constitute 52.77% of all sessions, micro-averaged across dialogues. This proportion does not alone characterize retrieval difficulty because most content in an evidence session may be irrelevant to a particular question.

Questions. As described above, the question typology is based on three axes and covers different semantic question types. The dataset consists of both open-ended and closed-ended questions. The question-level timestamp is of equal importance to the session-level timestamp, given that the correct interpretation of the former is critical to generating an accurate response.

Dialogues Characters. The dialogue characters represent diverse behavioral scenarios, which increases the realism of the user–assistant interactions and reflects authentic user behavior. The dialogues also incorporate sociocultural features characteristic of the Russian language and culture. Information about the characters and topics is presented in Subsection B.2 of the Appendix.

Feature	Value / Description
Total dialogues	85
Total questions	1,543
Average dialogue length	≈ 340,000 characters
Utterances per dialogue	180 – 998
Session definition	Single interaction with the assistant on a specific day
Sessions per dialogue	12 – 85
Evidence sessions	52.77% of all sessions (micro-average)
Questions per dialogue	14 – 22
Number of Question Types	17 base types
Language	Contemporary standard Russian (user specific)
Dialogue specifics	Include language errors Reflect Russian sociolinguistic worldview
Offensive content	None (no offensive, insulting, or threatening data)

Note: Dialogues are independent of each other.

Table 3: RUMBA Dataset Features and statistics.

3.4 Dataset creation

Russian version. The dataset was developed in multiple stages, including taxonomy design and dialogue construction. Initially, information extraction questions — constituting the majority of the dataset — were created, subsequently, reasoning questions and a category of temporal questions were incorporated.

The dialogue creation process involved a total of 26 contributors, comprising staff members and crowd workers (ABC Elementary[3] and Rambler&Co companies[4]). Each participant received monetary compensation according to the terms and conditions specified in their respective employment or contractual agreements. Information about the participants (gender, age, and field of expertise) is presented in Fig.7 of Appendix.

Data were collected over 20 working days. The process comprised two main stages: the creation of dialogues and associated questions by the authors, followed by a validation phase conducted by other contributors to ensure quality and consistency. More detailed information about the validation process can be found in Section B of Appendix.

[3]https://elementary.center

[4]https://rambler-co.ru

Language	Dialogue text	+ roles	+ answer prompt
RU	341,103.13	344,758	346,751
EN	320,782.08	324,437	326,430

Table 4: Context lengths in the Russian and English versions of RUMBA, measured in characters.

Before the start of data collection, all participants received comprehensive instructions (See Fig. 8 in Appendix), were acquainted with the question taxonomy, and were provided with the matrix.

The dialogues were created in real time. User utterances were written by human participants, while GigaChat LLM was selected to generate the assistant's responses, as it is specifically designed for the Russian language and Russian-speaking users, and is the only model pretrained on Russian data (Mamedov et al., 2025). The dialogue authors used the web version of GigaChat Max dated July 1–31 and October 21–November 10, 2025.

English version. We additionally provide an aligned English diagnostic split. Given the large size of the dialogue data, controlled manual translation would be too resource-intensive. Therefore, the English version of the dataset was produced using automated LLM translation, followed by human verification of the translation quality. This follows common practice in machine translation evaluation and multilingual benchmark construction, where full human assessment is costly and machine-translated data is often validated or post-edited on selected subsets (Lo and Knowles, 2023; Singh et al., 2025b; Rajaee et al., 2025).

For the assessment of translation quality, 10% of the translated dataset (i.e., 9 dialogues or a total of 185 sessions) was uniformly sampled and evaluated by human annotators. Translation quality was assessed using the *Critical* and *Task-specific* criteria defined in POLLUX (Martynov et al., 2025), with results summarized in Table 8. All 3,543 translated pairs RU-utterance — EN-utterance of the 9 dialogues were validated by domain-qualified specialists. Each evaluator compared the original Russian text with its English translation according to the POLLUX criteria, and every RU-EN pair was assessed by three annotators. The overall translation quality score was estimated at 0.88 (Table 9).

The next stage involved improving the quality of the translated set through a careful validation of the question–answer pairs and the corresponding evidence sessions. A total of 1,543 question–answer pairs (3,086 utterances in total) and 2,232 evidence sessions containing responses to the questions were each validated by three annotators. Where necessary, corrections were made to the English versions of the question–answer pairs and sessions. This helped preserve the logical and semantic coherence of the question, answer and evidence sessions to the same extent as in the original Russian version. More details on translation and its validation are provided in Appendix C.

4 Evaluation

4.1 Baselines

We evaluate two families of baselines that correspond to two common ways of using long interaction histories in LLM-based assistants: full-context inference and memory-based retrieval-augmented inference.

Full-context baselines. The first family represents long-context models that receive the complete dialogue history directly in the input context and generate the answer from this full evidence. This setting provides a classic upper-bound-style baseline for memory use. It also tests whether the model can locate and integrate relevant evidence in very long conversational contexts. In our experiments, we selected models with context windows around or above 1M tokens: fitting the largest dialogue requires context lengths roughly above 400K tokens, while the next practical tier among available long-context models is around 1M tokens. This constraint excluded Russian-oriented models, for which we did not find suitable context lengths, as well as available open-source models from the `Qwen` and `DeepSeek` families. Although `Qwen3.5-Flash` and `Qwen3.5-Plus` nominally satisfy the context-length requirement, provider-side content inspection rejected benchmark inputs with `DataInspectionFailed` errors in our setup, so we excluded them from the final evaluation. Prior long-term memory benchmarks similarly use long-context LLMs as a natural comparison point for evaluating implicit memory mechanisms (Maharana et al., 2024; Wu et al., 2025). We therefore evaluated both proprietary and open-source long-context models. The proprietary models are `gpt-4.1-mini`, `gpt-5.4`, `gemini-3.1-flash-lite`, `grok-4.1-fast`, and `claude-sonnet-4.6`; the open-weight models are `llama-4-maverick` and `minimax-01`.

Memory Agent/RAG baselines. The second family represents retrieval-based memory systems (Lewis et al., 2020). Unlike full-context baselines, these methods construct a persistent memory representation from the dialogue history during ingestion stage. At question-answering time, a small set of relevant memories is retrieved and inserted into the answer prompt. The final answer is generated by the same fixed answering model, `openai/gpt-4.1-mini`, across all memory-based baselines. Thus, whenever we refer to an Agent/RAG or memory-based method, we compare memory layers rather than answer-generation models. Performance differences within this family primarily reflect how well each method stores, updates, structures, and retrieves memories for the fixed answering model.

Our simple RAG baseline stores original message-pair chunks from the dialogue history and retrieves them directly from a vector database. The other systems go beyond this plain setup: they use LLM-based components to extract, transform, consolidate, or structure memories before retrieval. We therefore use Agent/RAG as a compact label for retrieval-based memory systems with different degrees of LLM-mediated memory processing, even though not all of them are agents in the narrow sense. (Singh et al., 2025a; Wang et al., 2024).

In our experiments, this family includes simple vector-based RAG as well as dedicated memory frameworks: `mem0` (vector-based) and `mem0g` (graph-based) (Chhikara et al., 2025), `graphiti` (open-source `zep`) (Rasmussen et al., 2025), `cortex` (Bhattacharjee, 2025), and `memOS` (Li et al., 2025). We followed the setup described in the corresponding papers and repositories. For reproducibility, we integrated the open-source frameworks as `git` submodules, while `mem0(g)` was accessed through its hosted commercial API. When a framework allowed replacing the embedding backend, we used language-specific embedding models of comparable dimension and parameter scale: `FRIDA` (ai-forever, 2024) for Russian and `Qwen3-Embedding-0.6B` (Zhang et al., 2025) for English, selected as strong retrieval-oriented embedding backbones on MTEB benchmark (Muennighoff et al., 2023).

4.2 Evaluation Pipeline

In addition to the dataset, we provide an evaluation methodology and an accompanying pipeline (Fig.10) as a reference protocol for validating memory-based systems. The proposed pipeline follows evaluation practices commonly used in long-context dialogue and memory benchmarks referenced in Sec.2.

RUMBA is evaluated using a two-stage pipeline: ***add*** and ***eval***. In the first *add* stage, the memory system processes the dialogue data sequentially as user–assistant pairs and populates its internal storage by extracting and storing memories (along with their timestamps) from the conversations. Dialogues belonging to different users are processed independently, and the resulting memories are stored in isolated user-specific memory spaces, e.g., database collections. The Russian and English subsets are ingested separately, resulting in distinct language-specific memory stores. This stage is skipped in the full-context evaluation setup.

In the *eval* stage, the system is evaluated using the open-source `Lighteval` framework (Habib et al., 2023). For each benchmark question, the system first retrieves the top-$k = 10$ relevant memories from the corresponding store. The retrieved timestamped[5] memories, along with timestamped question, are then provided as context to an answer-generation model, which produces a response based on a fixed language-specific prompt (Fig. 11). The generated answer is evaluated using an LLM-as-Judge procedure that compares the predicted answer with the reference answer. Different judge models are used depending on the evaluation language. For Russian evaluation, we use `POLLUX` (Martynov et al., 2025), a pretrained judge model for Russian language, which was selected based on a reported correlation of $\rho = 0.704$ with expert judgments. For English evaluation, the same scale and criteria are applied, while `DeepSeek-R1` is used as a judge model. We discuss the choice of judges in Appendix E.1. The pipeline produces LLM-as-Judge accuracy as the primary metric, with F1 reported as a complementary one. The judge assigns labels 0–3; labels 0 and 1 are mapped to incorrect, while labels 2 and 3 are mapped to correct for binary accuracy. F1 is the mean per-question bag-of-words F1 between the generated and reference answers, using the built-in `Lighteval` implementation: words are split on whitespace and treated as sets, and the maximum F1 is used for questions with multiple references. The full evaluation criteria and judge prompt templates are provided in Appendix E.2.

The analysis includes: (1) a family-level comparison of retrieval-based memory systems against full-context baselines; and (2) a matched-answerer comparison between the `gpt-4.1-mini` full-context baseline and Agent/RAG systems; and (3) within-family slice contrasts over different benchmark scopes. Confidence intervals are estimated by bootstrap resampling over questions, and p-values are Holm-corrected within each comparison block. All results are reported separately for Russian and English as well. Because `mem0g` is atemporal in our setup, we exclude it from Agent/RAG family aggregates for temporal, temporal-expression, and scope–temporal analyses.

5 Results

We report descriptive overall scores for each method in Table 5 and then analyze performance across the main benchmark axes: session scope, temporal reasoning, temporal-expression explicitness, and semantic type. A detailed description of the slice-level analysis and additional results are provided in Appendix F. An example of a single model case study is provided in Appendix G.

With the answerer controlled, the `gpt-4.1-mini` **Full-context baseline exceeds the overall Agent/RAG** score by +4.28 points in Russian and +8.64 in English. Method-level results are mixed: cortex, memOS, and simple RAG score significantly higher in Russian; in English, memOS scores significantly higher, while simple RAG is not significantly different (Table 6).

Semantic supergroup analysis shows that the **observed Full-context advantage is concentrated in Reasoning and Extraction questions**, but not in Abstention. Across both Agent/RAG and Full-context systems, Extraction questions are significantly easier than Reasoning questions in both languages, with within-family accuracy gaps ranging from $+9.35$ to $+14.24$ points (Table 7).

[5]The `mem0g` API did not provide timestamps for graph memories. To evaluate the graph-based implementation as is, we did not augment it with timestamped vector memories, resulting in an atemporal graph-based setup.

Method		Russian		English	
		LLM-as-Judge	F1	LLM-as-Judge	F1
Agent/RAG	**Embedder**				
mem0g (atemporal)	–	34.93 ± 1.21	20.52 ± 0.81	35.26 ± 1.22	24.27 ± 0.83
graphiti (open-source zep)	FRIDA | Qwen3-Embedding-0.6B	42.71 ± 1.26	27.00 ± 0.91	45.11 ± 1.27	29.13 ± 0.90
mem0	–	53.21 ± 1.27	30.10 ± 0.95	54.12 ± 1.27	35.05 ± 0.95
cortex	FRIDA | Qwen3-Embedding-0.6B	68.24 ± 1.19	36.32 ± 0.99	55.80 ± 1.26	34.26 ± 0.94
simple RAG	FRIDA | Qwen3-Embedding-0.6B	$\mathbf{68.89 \pm 1.18}$	36.17 ± 0.98	65.46 ± 1.21	39.00 ± 0.96
memOS	FRIDA | Qwen3-Embedding-0.6B	66.82 ± 1.20	$\mathbf{38.27 \pm 1.00}$	$\mathbf{69.99 \pm 1.17}$	$\mathbf{42.59 \pm 0.96}$
Full context	**Context size**				
llama-4-maverick	1M tokens	48.35 ± 1.27	27.18 ± 0.97	50.81 ± 1.27	28.58 ± 0.95
minimax/minimax-01	1M tokens	56.71 ± 1.26	31.99 ± 1.02	57.42 ± 1.26	31.68 ± 0.90
gpt4.1-mini	1M tokens	60.08 ± 1.25	27.26 ± 0.87	62.93 ± 1.23	33.68 ± 0.89
gemini-3.1-flash-lite	1M tokens	70.06 ± 1.17	39.30 ± 1.06	69.09 ± 1.18	39.93 ± 0.97
x-ai/grok-4.1-fast	2M tokens	76.22 ± 1.08	34.40 ± 0.92	79.59 ± 1.03	23.34 ± 0.76
claude-sonnet-4.6	1M tokens	78.61 ± 1.04	$\mathbf{43.03 \pm 1.02}$	81.98 ± 0.98	$\mathbf{40.98 \pm 1.01}$
gpt-5.4	1.05M tokens	$\mathbf{83.60 \pm 0.94}$	40.43 ± 0.95	$\mathbf{83.99 \pm 0.93}$	40.75 ± 0.90

Table 5: Overall RUMBA results for retrieval-based memory systems and full-context baselines. All scores are reported on a 0–100 scale; uncertainty is reported as standard error.

Agent/RAG	Lang.	GPT-4.1-mini FC	Agent/RAG	Δ [95% CI]
cortex	RU	60.08	68.24	-8.17 [-10.89, -5.44]
cortex	EN	62.93	55.80	+7.13 [4.41, 9.85]
mem0	RU	60.08	53.21	+6.87 [4.02, 9.66]
mem0	EN	62.93	54.12	+8.81 [6.03, 11.60]
mem0g	RU	60.08	34.93	+25.15 [22.29, 28.06]
mem0g	EN	62.93	35.26	+27.67 [24.89, 30.46]
memOS	RU	60.08	66.82	-6.74 [-9.46, -4.08]
memOS	EN	62.93	69.99	-7.06 [-9.66, -4.47]
simple RAG	RU	60.08	68.89	-8.81 [-11.47, -6.22]
simple RAG	EN	62.93	65.46	-2.53 [-5.25, 0.13]
graphiti (zep)	RU	60.08	42.71	+17.37 [14.52, 20.22]
graphiti (zep)	EN	62.93	45.11	+17.82 [14.91, 20.74]
Overall	RU	60.08	55.80	**+4.28** [2.08, 6.43]
Overall	EN	62.93	54.29	**+8.64** [6.42, 10.77]

Table 6: Direct matched-answerer comparison of the `gpt-4.1-mini` Full-context baseline with each Agent/RAG system and the overall Agent/RAG family mean. Scores and differences are reported on a 0–100 accuracy-point scale, with Δ = `gpt-4.1-mini` FC − Agent/RAG; confidence intervals are paired bootstrap 95% CIs over the same 1,543 questions.

Contrast	Family / Language	A	B	Δ [95% CI]
Single-session − Multi-session	Agent/RAG Russian	63.92	43.18	+20.73 [17.71, 23.83]
Single-session − Multi-session	Agent/RAG English	62.92	40.87	+22.06 [19.02, 25.14]
Single-session − Multi-session	Full context Russian	76.65	53.69	**+22.96** [20.07, 25.88]
Single-session − Multi-session	Full context English	78.21	55.70	**+22.51** [19.57, 25.50]
Atemporal − Temporal	Agent/RAG English	60.02	52.87	**+7.15** [3.54, 10.70]
Atemporal − Temporal	Full context Russian	69.39	62.96	+6.43 [3.10, 9.67]
Atemporal − Temporal	Full context English	70.91	65.30	+5.61 [2.23, 8.93]
Explicit time evidence − Implicit time evidence	Agent/RAG Russian	63.33	44.35	+18.99 [9.68, 28.61]
Explicit time evidence − Implicit time evidence	Agent/RAG English	60.20	38.55	+21.65 [11.82, 31.34]
Explicit time evidence − Implicit time evidence	Full context Russian	76.19	46.17	**+30.02** [21.90, 37.77]
Explicit time evidence − Implicit time evidence	Full context English	76.89	47.00	**+29.89** [21.61, 38.35]
Extraction − Reasoning	Agent/RAG Russian	56.12	41.88	**+14.24** [10.26, 18.11]
Extraction − Reasoning	Agent/RAG English	53.82	41.67	+12.15 [8.32, 16.02]
Extraction − Reasoning	Full context Russian	69.20	55.02	**+14.18** [10.22, 18.19]
Extraction − Reasoning	Full context English	69.85	60.51	+9.35 [5.48, 13.33]

Note: Agent/RAG temporal contrasts exclude `mem0g`.

Table 7: Main statistically significant within-family slice-difficulty contrasts. Scores are family-level means of binary LLM-as-judge correctness across methods within each family; scores and differences are reported on a 0–100 accuracy-point scale.

Among the three main benchmark axes, session scope is the strongest and most consistent difficulty driver. Across both method families and both languages, single-session questions are easier than multi-session questions by 20.73–22.96 accuracy points, with all confidence intervals excluding zero (Table 7). Full-context systems stay more accurate on multi-session questions but are not immune to this difficulty.

Temporal questions are also harder on average, although the effect is smaller and less uniform than the single-to-multi degradation. Among statistically significant contrasts, the atemporal-to-temporal gap ranges from 5.61 to 7.15 accuracy points depending on language and method family; the Russian Agent/RAG contrast is not significant after correction (Table 7). In English, the Full-context–Agent/RAG gap is +12.43 on temporal and +10.89 on atemporal questions.

Implicit temporal expressions are a particularly strong failure mode. Questions whose supporting evidence contains explicit temporal expressions are substantially easier for models than questions whose evidence requires resolving implicit temporal expressions, for both method families and both languages (Table 7). The gap is especially large for Full-context systems: +30.02 accuracy points in Russian and +29.89 accuracy points in English.

6 Conclusion

We introduced RUMBA, the first benchmark for evaluating long-term conversational memory in Russian, together with an aligned English version. Unlike existing English-centric benchmarks, RUMBA accounts for the cultural and linguistic specifics of Russian, including natural interaction patterns and temporal expressions. We proposed a multilingual methodology with a fine-grained taxonomy covering contextual recall, entity tracking, temporal reasoning, and multi-step integration of evidence. We released RUMBA as an open-source resource[6] and evaluated state-of-the-art LLMs under both RAG and full-context scenarios. Our results show that current models struggle significantly with tasks requiring simultaneous memory retention, temporal reasoning, and multi-step integration in Russian. RUMBA thus serves as both a diagnostic tool and a catalyst for developing culturally aware conversational agents with robust long-term memory. Future work includes scaling the dataset, enriching reasoning-heavy examples, and probing internal memory representations beyond QA.

Limitations

Data. The dataset size, while adequate for initial benchmarking, is not large enough to introduce substantial difficulties for very long-context models. Scaling up both the number of dialogues and their length would further stress-test long-context capabilities. In addition, extraction-oriented questions dominate, whereas complex reasoning and temporally entangled queries are underrepresented. This imbalance reflects practical considerations and ensures a solid foundation, but it limits evaluation of higher-order memory abilities. Finally, the English version is obtained through automatic translation rather than manual rewriting, with no explicit localization applied. This may make the English split less culturally natural and introduce translation artifacts that can affect model understanding and embedding-based retrieval.

Evaluation. Our Russian and English evaluations use different LLM judges. Consequently, part of the observed cross-lingual difference may stem from judge-specific calibration rather than model performance alone (Thakur et al., 2025; Fu and Liu, 2025). RUMBA also follows a standard QA pipeline, which enables transparent and reproducible evaluation of output correctness but does not directly probe the model's internal memory representations. Nevertheless, the benchmark targets memory capabilities by construction: unlike open-domain QA, RUMBA centers on user-specific information that evolves over time, including repeated updates, corrections, and changes in personal facts.

Retrieval diagnostics. Our experiments include retrieval-based memory systems, but do not provide a systematic diagnostic analysis of retrieval design choices, such as the choice of embedder, ranking method, retrieval depth, etc. This is outside the main scope of the paper: our goal is to compare memory approaches under a unified pipeline and to show how RUMBA can be used for slice-level diagnostics, rather than to optimize a particular retrieval architecture.

Future work. The rapid development of long-term memory benchmarks and memory-augmented agents suggests several natural directions for future work. Recent benchmark extensions, such as LongMemEval-V2 (Wu et al., 2026), and ongoing updates to memory frameworks[7], appeared after RUMBA creation, illustrate the increasing importance of larger-scale, more agentic, and more diagnostic memory evaluation. Looking ahead, future versions of RUMBA can (i) scale the dataset by adding more dialogues and extending session lengths, (ii) enrich the benchmark with a broader variety of reasoning-intensive and temporally entangled examples to improve class diversity, (iii) develop complementary diagnostics for retrieval behavior, memory retention, consistency, and update tracking across sessions, thereby further strengthening RUMBA's utility as a comprehensive evaluation framework.

Ethical consideration

Data collection and annotation All dialogues, questions, and answers were collected from scratch, without reusing any existing copyrighted or restricted datasets. Annotations were performed by native Russian speakers with prior experience in linguistic tasks. All annotators were compensated at a rate exceeding the local minimum wage, and participation was voluntary. Written informed consent was obtained from all annotators, and no personally identifiable information was collected.

Licensing and accessibility The resulting benchmark is released under a public license (MIT). The Russian and English subsets are both distributed under the same terms.

AI assistant usage During the preparation of this manuscript, the authors used AI assistants solely for editing and improving the English language of the text, including grammar correction, phrasing refinement, and vocabulary suggestions. All intellectual contributions, benchmark design decisions,

[6] https://github.com/ai-forever/RUMBA

[7] https://mem0.ai/research

and interpretations remain the sole responsibility of the authors.

Acknowledgments

We thank all those who contributed to the RUMBA dataset. The dialogues were written, edited, and reviewed by Irina Shamova, Ksenia Kuleshova, Olga Borodina, Maria Avilova, Angelina Pryadko, Elena Dorofeeva, Tatiana Chabanyuk, Anna Plotnikova, Margarita Khurramova, Maksim Anoshin, Pavel Prosyanik, Ksenia Kondrakova, Yulia Piliguzova, Ekaterina Kuryachaya, Allam Atabaev, Inessa Kurevleva, Zakhar Chaichenko, Victoria Nikolaeva, Olga Borisova, Vsevolod Alipov, Danil Vyazovov, Evgenia Efremova, Timur Shenturk, and Konstantin Gridin. Final validation was performed by Olga Borisova, Taisiya Metelkina, Maria Saburbaeva, Yulia Piliguzova, Ekaterina Kuryachaya, and Allam Atabaev, and we are especially grateful to Artemy Stankevich for making the validation process possible. The English version was worked on by Ekaterina Kuryachaya, Allam Atabaev, Natalia Chukicheva, Ksenia Kuleshova, Dmitry Kosourov, Eleonora Petrikova, Alena Gubar, Maria Petrova, Lada Kovalchuk, and Daria Panaiotti. Finally, we appreciate the management efforts of Elvira Budaeva, Pavel Kovalev, Anna Kostikova, and Dmitry Bocharov, as well as the overall coordination by Yuliya Vasilieva.

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A Taxonomy details

Semantic axis. Questions are broadly classified into information-extraction and reasoning questions, based on the information available about the user. A detailed classification of question types, along with their descriptions and illustrative examples, is presented in Table 26.

Quantitative axis. Questions are classified into two types. *Single-session questions*: the answer to the question is contained within a single session, whereas in *multi-session questions*, the answer is distributed across multiple sessions. The distribution of questions on the quantitative axis is depicted in Fig.4.

Axis of temporality. Questions are classified as *atemporal*, for which comprehension of the dialogue's temporal framework is not required, or *temporal*, for which such temporal understanding is essential to answer. The distribution of questions along axis of temporality is presented in Fig.5. The taxonomy of temporal questions is presented in Fig.3.

Each temporal question is accompanied by a tag indicating how temporal information was presented in the dialogue: *explicit temporal expression* (time was indicated explicitly, e.g., a specific day and/or month, year, etc.), *implicit temporal expression* (time was indicated implicitly, e.g., with words *yesterday, 2 days ago*, etc.), or *no temporal expression* (no temporal information was provided). Fig.6 presents the distribution of temporal expression tags.

B Annotation details

B.1 Dataset Creation

On average, the development (creation and validation) of a single dialogue and its corresponding set of questions required approximately 2 working days to complete.

To standardize the dialogue creation process, a matrix was developed containing the required question types for each dialogue. Authors used this matrix during dialogue construction, which ensured diversity of question types within each dialogue.

Dialogue Validation Process. The validation process consisted of three stages. (1) Authors cross-validated dialogues, checking question adequacy, taxonomy alignment, and consistency of dialogue and question dates. Necessary corrections to questions and dates were introduced at this stage. (2) Questions and corresponding sessions were validated using an LLM (see the paragraph below *Evidence annotation*). Based on LLM answers, validators — including the seven highest-performing authors and three independent validators — evaluated whether (a) the question matched its taxonomy category; (b) all dates were explicit and temporally consistent; (c) the answer was contained in the designated sessions and absent from unrelated ones; and (d) the dialogue was internally consistent and provided sufficient information to answer the question. Validators also revised questions and sessions when necessary and occasionally added new sessions. (3) Final validation was performed after all revisions. Each dialogue was reviewed by three validators who had not previously reviewed that dialogue.

Finally, the complete set of question–answer pairs, session dates, question dates, and taxonomy labels was independently validated by human annotators with an overlap of three.

Evidence annotation. We additionally constructed an utterance-level evidence annotation to identify where answer-relevant information appears in the dialogues. The input consisted of spreadsheet files with written dialogues, where each row contained a speaker role, an utterance, and a session-boundary marker, together with question-level metadata: the question, its reference answer, and its type. We first reconstructed session identifiers from the boundary markers and extracted the set of questions with their associated answers, types, and source sessions.

Then, for each utterance–question pair, we queried `GigaChat-Max` and asked whether the utterance contained explicit or implicit information relevant

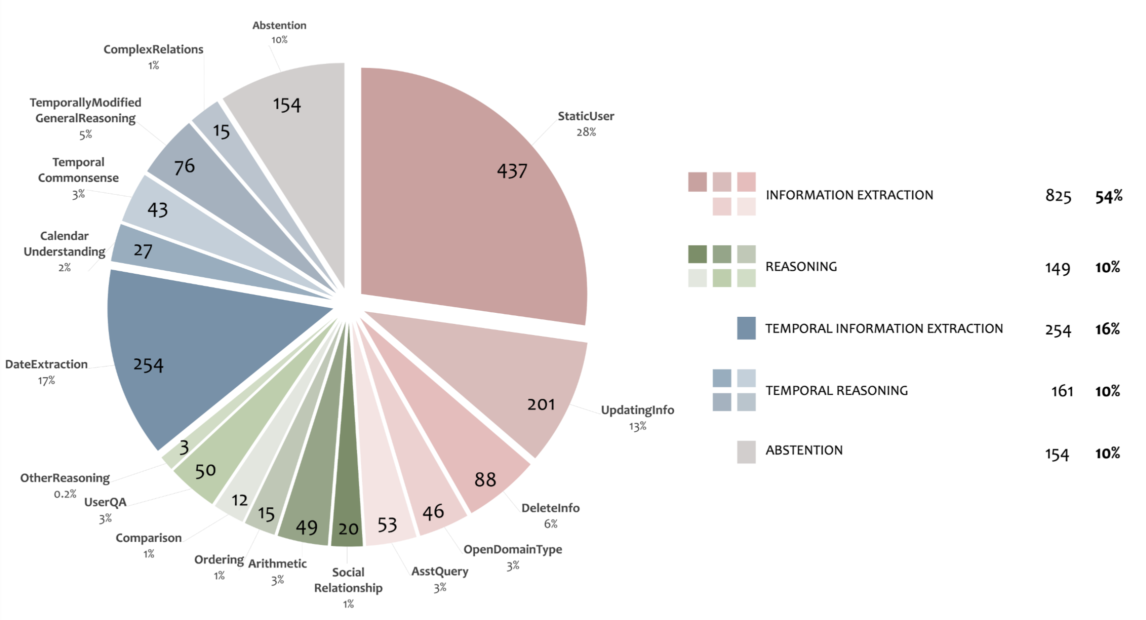


Figure 2: Overview of RUMBA Taxonomy.

Figure 3: Overview of temporal questions Taxonomy: distribution by type, with corresponding counts and percentages.

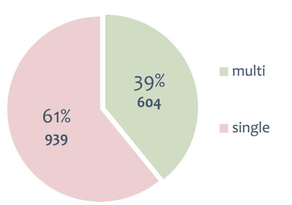


Figure 4: Distribution across the quantitative axis, reflecting the ratio between single-session and multi-session questions, including both the number of questions and their percentage representation.

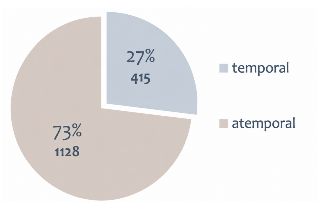


Figure 5: Distribution across the axis of temporality, reflecting the ratio between temporal and atemporal questions, including both the number of questions and their percentage representation.

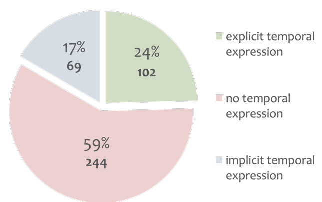


Figure 6: Temporal tags: distribution of explicit, implicit, non-expressed temporal expressions in the dialogues, including both the number of expressions and their percentage representation.

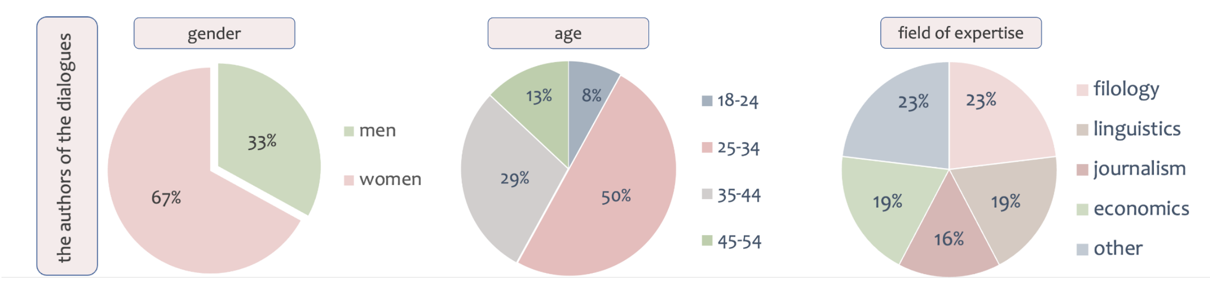
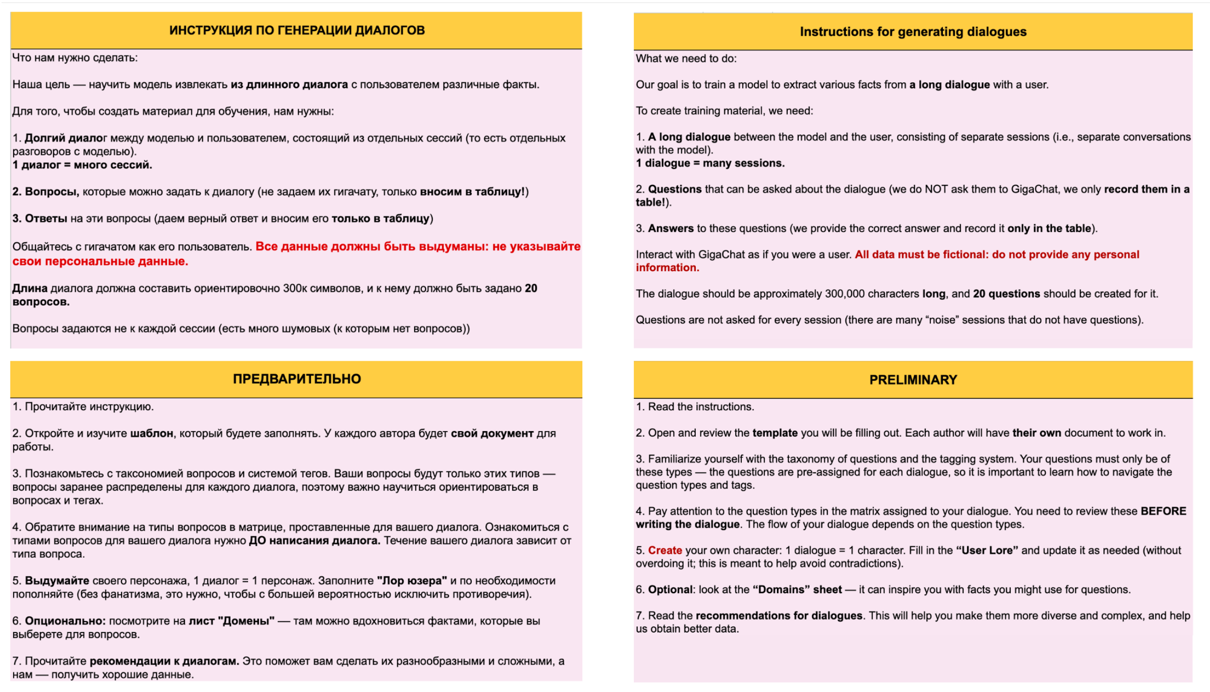


Figure 7: Information about the dialogue authors including gender, age and field of expertise distribution, reported in terms of percentages.



ИНСТРУКЦИЯ ПО ГЕНЕРАЦИИ ДИАЛОГОВ

Что нам нужно сделать:

Наша цель — научить модель извлекать **из длинного диалога** с пользователем различные факты.

Для того, чтобы создать материал для обучения, нам нужны:

1. **Долгий диало**г между моделью и пользователем, состоящий из отдельных сессий (то есть отдельных разговоров с моделью).
1 диалог = много сессий.

2. Вопросы, которые можно задать к диалогу (не задаем их гигачату, только **вносим в таблицу!**)

3. Ответы на эти вопросы (даем верный ответ и вносим его **только в таблицу**)

Общайтесь с гигачатом как его пользователь. **Все данные должны быть выдуманы: не указывайте свои персональные данные.**

Длина диалога должна составить ориентировочно 300к символов, и к нему должно быть задано **20 вопросов.**

Вопросы задаются не к каждой сессии (есть много шумовых (к которым нет вопросов))

ПРЕДВАРИТЕЛЬНО

1. Прочитайте инструкцию.

2. Откройте и изучите **шаблон**, который будете заполнять. У каждого автора будет **свой документ** для работы.

3. Познакомьтесь с таксономией вопросов и системой тегов. Ваши вопросы будут только этих типов — вопросы заранее распределены для каждого диалога, поэтому важно научиться ориентироваться в вопросах и тегах.

4. Обратите внимание на типы вопросов в матрице, проставленные для вашего диалога. Ознакомиться с типами вопросов для вашего диалога нужно **ДО написания диалога.** Течение вашего диалога зависит от типа вопроса.

5. **Выдумайте** своего персонажа, 1 диалог = 1 персонаж. Заполните **"Лор юзера"** и по необходимости пополняйте (без фанатизма, это нужно, чтобы с большей вероятностью исключить противоречия).

6. **Опционально:** посмотрите на **лист "Домены"** — там можно вдохновиться фактами, которые вы выберете для вопросов.

7. Прочитайте **рекомендации к диалогам.** Это поможет вам сделать их разнообразными и сложными, а нам — получить хорошие данные.

Instructions for generating dialogues

What we need to do:

Our goal is to train a model to extract various facts from **a long dialogue** with a user.

To create training material, we need:

1. **A long dialogue** between the model and the user, consisting of separate sessions (i.e., separate conversations with the model).
1 dialogue = many sessions.

2. **Questions** that can be asked about the dialogue (we do NOT ask them to GigaChat, we only **record them in a table!**).

3. **Answers** to these questions (we provide the correct answer and record it **only in the table**).

Interact with GigaChat as if you were a user. **All data must be fictional: do not provide any personal information.**

The dialogue should be approximately 300,000 characters **long**, and **20 questions** should be created for it.

Questions are not asked for every session (there are many “noise” sessions that do not have questions).

PRELIMINARY

1. Read the instructions.

2. Open and review the **template** you will be filling out. Each author will have **their own** document to work in.

3. Familiarize yourself with the taxonomy of questions and the tagging system. Your questions must only be of these types — the questions are pre-assigned for each dialogue, so it is important to learn how to navigate the question types and tags.

4. Pay attention to the question types in the matrix assigned to your dialogue. You need to review these **BEFORE writing the dialogue**. The flow of your dialogue depends on the question types.

5. **Create** your own character: 1 dialogue = 1 character. Fill in the **“User Lore”** and update it as needed (without overdoing it; this is meant to help avoid contradictions).

6. **Optional**: look at the **“Domains” sheet** — it can inspire you with facts you might use for questions.

7. Read the **recommendations for dialogues**. This will help you make them more diverse and complex, and help us obtain better data.

Figure 8: Author instructions describing the required manuscript components and the requirement to avoid personal data in dialogues.

to answering the question. If no relevant information was present, the model returned `None`; otherwise, it returned a concise excerpt or comment describing the relevant evidence.

The output was an augmented spreadsheet for each dialogue. It preserved the `session_id`, speaker role, and utterance text, and added one evidence column per question. Each evidence column encoded the question, question type, reference answer, and associated source sessions in its header, while its cells contained the model-produced evidence for each utterance or an empty value when no evidence was detected. This produced a dense utterance-by-question evidence map for subsequent human inspection and diagnostic analysis.

The LLM validation prompt for GigaChat Max in Russian (and translated into English) is shown in Fig. 9.

B.2 Users characters

Dialogues Characters. Users characters represented in the dialogues are heterogeneous and multifaceted, and their interactions encompass a range of thematic domains, which enhances the realism of the constructed interactions and brings them closer to authentic user–assistant conversations. The behavioral scenarios assigned to these characters likewise reflect real-world user patterns, ranging from memory-related operations (such as storing, forgetting, and modifying information) to various strategies for expressing intent. These include explicit requests to memorize a fact, introducing a fact within a single utterance or anaphorically, and linking facts across different sessions, etc. Information about the characters and topics is presented in

```
Тебе будет дан вопрос и диалог между пользователем и
ассистентом. Тебе нужно понять, есть ли в этом диалоге
информация,
которая может быть связана с вопросом.

Если такой информации нет, ответь None.

Если есть информация, связанная с вопросом, напиши кратко
соответствующий отрывок/отрывки из диалога,
где эта информация упоминается.

Заметь, что информация может быть в неявном виде, нужно
распознавать ее. Уделяй внимание мельчайшим деталям. Может
пользователь
незаметно упомянул то, что может быть каким-то образом связано с
вопросом.
Никаких лишних комментариев не нужно.

Диалог: {session}
Вопрос: {question}
```

```
You will be given a question and a dialogue between the user and
the assistant. You need to determine whether this dialogue
contains information
that may be related to the question.

If there is no such information, answer None.

If there is information related to the question, briefly write
the relevant excerpt/excerpts from the dialogue
where this information is mentioned.

Note that the information may be implicit, and you need to
recognize it. Pay attention to the smallest details. The user
may have subtly
mentioned something that could in some way be related to the
question.
No extra comments are needed.

Dialogue: {session}
Question: {question}
```

Figure 9: Evidence annotation prompt template used during validation of dialogue sessions and their associated questions. The original Russian prompt is shown together with an English translation provided for visualization in the paper.

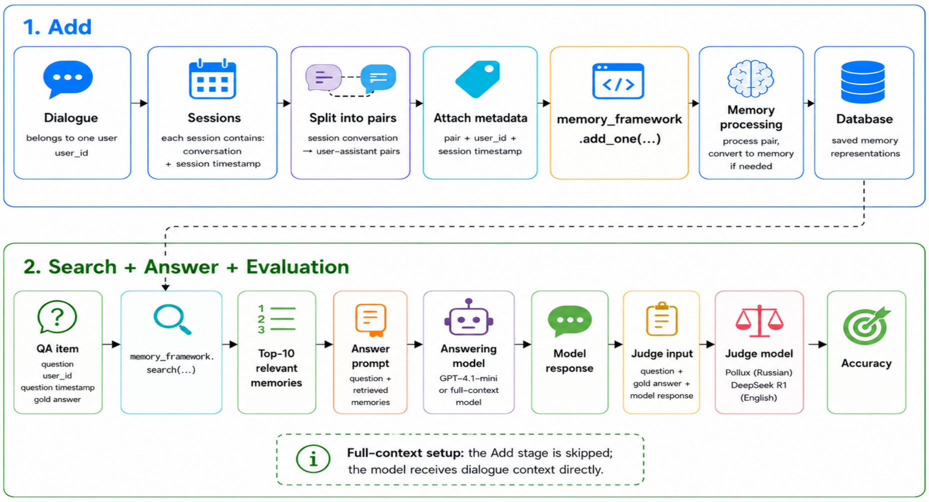


Figure 10: RUMBA validation pipeline.

Fig.12.

Sociolinguistic Portrait of the Characters. The dialogues reflect sociocultural realities characteristic of Russia and Russian culture, as they include grammatical constructions typical of Russian, along with culture-specific references such as idioms, proverbs, folk sayings, anecdotes, and allusions to canonical and popular Russian literary and cultural works. Dialogues intentionally include spelling, grammar, lexical, stylistic errors to approximate authentic user-generated input.

C Translation

Two translation granularities were used. Questions and answers were translated independently at the sentence level. Dialogue messages were translated at the session level, where the full sequence of messages within a session was provided to the model in a single request to preserve conversational context and message ordering. `gpt-4.1-mini`[8] was used as a translator model. Fig. 13 reports the system prompts and compact user-message schemas used for machine translation.

All remaining dataset fields were carried over unchanged, as they were either originally in English or language-independent.

C.1 Translation validation details

The translation validation was carried out by specialists who possess relevant academic qualifications in the field (civil law contract workers and crowd workers (ABC Elementary[9] and Rambler&Co companies[10]).

[8] https://openai.com/index/gpt-4-1/
[9] https://elementary.center
[10] https://rambler-co.ru

```
# КОНТЕКСТ:
У тебя есть доступ к воспоминаниям об одном пользователе из его разговоров с
ассистентом. Воспоминания содержат факты и временные метки, которые могут быть
релевантны для ответа.
Вопрос задан от лица пользователя и всегда относится именно к этому
пользователю. Вопрос задан в определенный момент времени.

# ИНСТРУКЦИИ:
1. Внимательно проанализируй все предоставленные воспоминания.
2. Учитывай временные метки воспоминаний.
3. Если информация противоречива, приоритет у самого недавнего воспоминания.
4. Если в воспоминаниях есть относительные временные отсылки («в прошлом году»,
«два месяца назад» и т. п.), преобразуй их в конкретные даты, месяцы или годы на
основе временной метки воспоминания.
5. Отвечай только на основе воспоминаний.
6. Не путай пользователя с другими людьми и персонажами, упомянутыми в
воспоминаниях.

# ФОРМАТ ОТВЕТА:
1. Ответ должен быть коротким (насколько это возможно из вопроса) и содержать
только недостающую часть, которую спрашивают.
2. Не пересказывай вопрос и не добавляй лишних пояснений.
3. Если вопрос предполагает ответ «да» или «нет», отвечай только: «да» или
«нет».
4. Если вопрос альтернативный («или»), отвечай только корректным вариантом из
вопроса. Пример: «Я люблю петь или танцевать?» → «Танцевать».
5. Если вопрос про день/дату или месяц, отвечай с годом (кроме случаев, где это
нельзя определить).
6. Если информации для ответа нет или она была удалена пользователем, ответ:
«Нет такой информации».
7. Ответ всегда должен быть на русском языке.

# ПОДХОД:
1. Внимательно изучи все воспоминания, содержащие информацию, связанную с
вопросом.
2. Проверь их содержание и временные метки.
3. Определи точный ответ по фактам из воспоминаний.
4. Убедись, что ответ короткий, конкретный и без относительных временных
формулировок.

Воспоминания для пользователя:
{% for memory in memories %}
{{ memory }}
{% endfor %}
Вопрос: {{question}}

Временная метка вопроса (текущая дата): {{query_date}}, день недели:
{{query_weekday}}

Ответ:
```

```
# CONTEXT:
You have access to memories about one user from their conversations with the
assistant. The memories contain facts and timestamps that may be relevant for
answering the question.
The question is asked from the user's perspective and always refers specifically
to this user. The question is asked at a specific point in time.

# INSTRUCTIONS:
1. Carefully analyze all provided memories.
2. Take the timestamps of the memories into account.
3. If the information is contradictory, prioritize the most recent memory.
4. If the memories contain relative time references ("last year", "two months
ago", etc.), convert them into specific dates, months, or years based on the
timestamp of the memory.
5. Answer only based on the memories.
6. Do not confuse the user with other people or characters mentioned in the
memories.

# ANSWER FORMAT:
1. The answer must be short (as short as the question allows) and contain only
the missing part being asked for.
2. Do not restate the question or add unnecessary explanations.
3. If the question requires a "yes" or "no" answer, respond only with: "yes" or
"no".
4. If the question is alternative ("or"), respond only with the correct option
from the question. Example: "Do I like singing or dancing?" → "Dancing".
5. If the question is about a day/date or month, include the year in your answer
(except in cases where it cannot be determined).
6. If there is no information to answer the question, or the information was
deleted by the user, respond: "No such information".
7. The answer must always be in English.

# APPROACH:
1. Carefully review all memories that contain information related to the
question.
2. Check their content and timestamps.
3. Determine the exact answer based on the facts in the memories.
4. Make sure the answer is short, specific, and does not contain relative time
references.

Memories for user:
{% for memory in memories %}
{{ memory }}
{% endfor %}
Question: {{question}}

Question timestamp (current date): {{query_date}}, weekday: {{query_weekday}}

Answer:
```

Figure 11: Answer-generation prompt templates used for Russian and English evaluation. The templates condition the retrieved memories (full dialogues in case of full context evaluation), memory timestamps, the benchmark question, and the question timestamp with its weekday.

Figure 12: High-level information about RUMBA dialogues. Dialogue characters include male and female adults as well as children. The top 10 conversation topics range from health to music and cinema. The dialogues also extensively reflect Russian discourse, including Russian history and art, social realities, notable figures, and broader societal characteristics.

Validation of translation quality: 10% of the translated dataset. Each evaluator received the original Russian text and its English translation for comparison and assessment of translation quality. Evaluators were provided with translation quality assessment criteria implemented in POLLUX (Martynov et al., 2025). The names of the criteria, their descriptions, and the scoring system are presented in Table 8.

The average time required to complete the evaluation of a single RU-EN pair of utterances was approximately 3 minutes.

Question-Answer-Evidence Sessions Validation. Experts evaluated a triplet: the evidence sessions and associated question–answer pairs. At this stage, it was essential to ensure that: (1) the translated question preserves the meaning and intent of the original question; (2) the translated gold answer to the question is consistent with the original answer; (3) the translated gold answer corresponds to the translated evidence session, i.e., the session indeed contains the information required to answer the question; and (4) the translated question–answer–session triplet is free of errors and contradictions and remains logically and semantically coherent to the same extent as the original Russian version. All issues were corrected.

The average time required to complete the evaluation of the triplet was approximately 5 minutes.

Non-evidence sessions (irrelevant to questions, filler sessions) were not manually validated by experts. For these sessions, we assume a translation quality score of 0.88, which was obtained through 10%-dataset validation.

The evaluators received compensation for each completed task in accordance with the terms stipulated in their employment contracts. The validation procedure was carried out using the ABC Elementary [11] platform.

D Evaluation details

Fig. 10 shows the RUMBA evaluation pipeline. Language-specific answer-generation prompts are provided in Fig. 11.

[11] `https://elementary.center/`

E Judge details

E.1 English judge choice

Our goal was to select an English judge that could be used with the same rubric as `POLLUX`, so that the English scores would be produced under an evaluation procedure as close as possible to the Russian one. We based the selection on JudgeBench (Tan et al., 2025) and its Hugging Face leaderboard[12]. JudgeLRM additionally reports 78.67% agreement between `DeepSeek-R1` and human judgments on PandaLM (Chen et al., 2025). Higher-ranked alternatives were not suitable for this setup: several fine-tuned judge or reward models are designed for pairwise preference comparison rather than scoring a single answer against a reference; Prometheus-style judges rely on their own scoring scale; and prompted GPT/o-series judges were avoided because GPT-family models are included among our answer-generation baselines. We therefore selected `DeepSeek-R1` as a strong non-GPT prompted judge compatible with our rubric. We found no single judge with reported human alignment for both languages under comparable conditions and thus used language-specific judges.

E.2 Judge prompts and verdicts

To obtain comparable scores for the Russian and English subsets, we used the same `POLLUX`-style evaluation format in both languages. The only difference is that `POLLUX` was evaluated with the prompt template and response format it was trained for, while `DeepSeek-R1`, being a general-purpose model, required a short task description and a JSON output format. Full templates are presented in Fig. 14.

The judge assigned one of 4 rubric labels. This rubric set was selected empirically. Labels 0–2 correspond to the standard `POLLUX` rubrics for the *Answer Correctness* criterion: incorrect or missing answer, incomplete answer, and answer matching or equivalent to the reference answer. We added label 3 for a frequent borderline case in our benchmark: the model answers the question correctly in terms of the key meaning, but adds extra details that are not supported by the question or the reference answer and would require checking the original dialogue. For the final binary accuracy score, labels 0 and 1 were treated as incorrect, while labels 2

[12] `https://huggingface.co/spaces/ScalerLab/JudgeBench`

Table 8: Translation Evaluation Criteria, their descriptions, and the scoring system

Criteria name	**Description**	**Scores**
Critical criteria		
(a) Format violation or (b) Censor block	(a) This criterion evaluates whether it is necessary to further grade the LLM's output based on other criteria or the LLM's output is unreadable and cannot be graded. (b) The criterion evaluates whether this is the LLM's output or there is just a placeholder with the text "Unfortunately, generative models cannot discuss such topics" or similar standard text where the LLM avoids answering.	**(a) 0:** We read the model's response and it can be further evaluated based on other criteria. **1:** The model's response shows excessive looping, or there are many artifacts, or the model's response is entirely in a different language, or the model simply copied the user's query. It does not make sense to further evaluate the model's response. **(b) 0:** The censor did not trigger, the LLM's output can be evaluated further based on other criteria. **1:** The censor triggered, there is nothing to evaluate.
Task-specific criteria		
Original goal	This criterion evaluates whether the communicative purpose of the source text is preserved in the translation: for example, to inform, advertise, accuse, tease, and so on.	**0:** In the translation, the source text intent is completely lost. **1: (a)** The source text intent is fully preserved in the translation. **(b)** In the translation, the source text intent is preserved but slightly blurred (for example, the aim was to teasingly poke fun at the reader, but only the humorous effect remains without the mockery).
Original tone	This criterion evaluates whether the tone of the source text is preserved in the translation. It refers to that part of the expressive plan which consists of the connotatively marked elements of the source text. If events are described with irony, the translator should (at least partially) retain this irony. If the information in the source text is presented neutrally, the translator should convey the same tone. Conversely, if the tone of the source text is not neutral, the translation should (at least partially) convey this as well.(Source: S. V. Tyulenev "Translation Theory")	**0:** The emotional tone of the text was completely lost in the translation. **1: (a)** The LLM completely preserved the tone of the source text. **(b)** In general, the translation preserved the tone of the source text, but it is somewhat diluted; it could have been handled better.
Author viewpoint	This criterion evaluates whether the translation distorts the author's position. The translation should not contradict the author's position. The author's position may include the author's point of view (for example, they may be subtly criticizing a product between the lines); ideological stance (often, the ideological position can be inferred from indirect signs); degree of confidence/uncertainty (if the author speaks about something uncertainly, it should not become a statement in the translation), etc.	**0:** The author's position was completely distorted during the translation. **1: (a)** The author's position is completely preserved, and the LLM conveyed the subtext, if there was any. **(b)** The author's position is generally preserved, but there are some issues (for example,the LLM overlooked certain veiled messages).
Compliance with functional style	This criterion evaluates whether the functional style is preserved in the translation. When translating, the text should maintain its functional style: literary, journalistic, official business, scientific, or colloquial-everyday, including the linguistic features of each style.	**0:** The LLM completely changed the functional style. **1: (a)** The translated text fully corresponds to the functional style in which the source text is written. **(b)** The LLM retained the functional style, but some linguistic features not typical for this functional style appeared in the translation.
Language norms	This criterion evaluates the compliance with lexical, grammatical, syntactic and stylistic norms of the target language. The translation should "sound natural", meaning it should contain appropriate vocabulary and grammar, syntax, and stylistic devices characteristic of the target language.	**0:** The LLM made 3 or more errors in compliance with lexical, grammatical, syntactic and stylistic norms of the target language. **1: (a)** The LLM made no errors in compliance with lexical, grammatical, syntactic and stylistic norms of the target language. **(b)** The LLM made one or two errors in compliance with lexical, grammatical, syntactic and stylistic norms of the target language.
Factual accuracy	This criterion evaluates the accuracy of conveying the factual information presented in the source text. The LLM should not distort factual (precision) information from the source text. An exception is made for factual information that would definitely not be understood by a native speaker of the target language: in such cases, the factual information should be adapted to convey the tone of the source text and the author's intent.	**0:** The LLM made factual errors, incorrectly conveying precise information from the source text. **1: (a)** The LLM conveyed the facts and their related context/subtext completely accurately. **(b)** The LLM did not make any factual errors but made a slip related, for example, to the subtext concerning some phenomenon or event.

Table 9: Criterion-specific scores obtained from human evaluation of translation quality.

Criteria	Value
Critical criteria (Format violation or Censor block)	0.99
Task-specific criteria	
Original goal	0.99
Original tone	0.98
Author viewpoint	0.99
Compliance with functional style	0.99
Language norms	0.9
Factual accuracy	0.97
General score	0.88

```
QA-pair translation

SYSTEM:
You are a professional translator. Translate Russian to natural English.
You are given a question-answer pair. Translate both together.
Rules:
- Preserve exact meaning and the relationship between question and answer.
- Keep punctuation and sentence type where possible.
- Preserve grammatical cues affecting meaning (gender, number, etc.).
- Do not omit, generalize, or add information.
- Keep names, titles, and quoted text consistent.
- If the answer is an alphabetical list, reorder it to be correct in English while preserving the items.

USER:
{"question": question_ru, "answer": answer_ru}


Dialogue-session translation

SYSTEM:
You are a professional translator. Translate Russian to natural English.
This is a conversation session; messages are context-dependent.
Rules:
- Preserve messages order and meaning exactly.
- Preserve formatting inside each message text: keep newlines, markdown, bullets, code blocks, etc - if any.
- Translate ONLY the message text; do not change dia_id.
- Output must match the provided schema.

USER:
{"dia_id": dia_id, "messages": [{"speaker": speaker, "text": text_ru}, ...]}
```

Figure 13: Translation message templates used to construct the English split. System prompts are shown verbatim; user messages are shown as compact payload schemas.

```
### Задание для оценки:
{question}

### Эталонный ответ:
{gold}

### Ответ для оценки:
{answer}

### Критерий оценки:
Правильность ответа

### Шкала оценивания по критерию:
0: Дан неправильный ответ или ответ отсутствует.

1: Ответ модели неполный (не на все вопросы задания получен ответ, в
формулировке ответа отсутствует часть информации).

2: Ответ модели совпадает с эталонным или эквивалентен ему. Для
отсутствия информации допускаются разные формулировки, если ясно
указано, что данных/воспоминаний для ответа нет.

3: Ответ модели совпадает по ключевому смыслу с эталонным, но
содержит дополнительную информацию, не подтверждаемую вопросом или
эталонным ответом.
```

```
You are an evaluation judge.

Your task is to evaluate the model's answer strictly against the
reference answer, which serves as the ground truth, using the
provided criterion and scoring rubric.
Do not use external knowledge. Carefully compare the model's answer
with the reference answer and assign the most appropriate score
according to the rubric.

Return your result in JSON format:
{
    "verdict": <0|1|2|3>,
    "reasoning": "<short explanation>"
}

### Evaluation Task:
{question}

### Reference Answer:
{gold}

### Model's Answer to Evaluate:
{answer}

### Evaluation Criterion:
Answer Correctness

### Scoring Rubric:
0: The answer is incorrect or missing.

1: The model's answer is incomplete (not all parts of the task are
addressed, or some required information is missing).

2: The model's answer matches the reference answer or is equivalent
to it. For cases with no available information, different phrasings
are acceptable as long as it is clearly stated that the
information/memory is unavailable.

3: The model's answer matches the key meaning of the reference
answer but includes additional information not supported by the
question or the reference answer.
```

Figure 14: LLM-as-Judge prompt templates used for Russian and English evaluation. Placeholders indicate the inserted benchmark question, reference answer, and model answer.

and 3 were treated as correct.

E.3 Qualitative Analysis

Judge error type	POLLUX	DeepSeek-R1
Correct answer judged as over-informative	2 (2.6%)	16 (16.7%)
Correct answer judged as wrong	2 (2.6%)	1 (1.0%)
Correct answer judged as incomplete	19 (24.7%)	9 (9.4%)
Over-informative answer judged as correct	8 (10.4%)	3 (3.1%)
Over-informative answer judged as wrong	3 (3.9%)	0 (0.0%)
Over-informative answer judged as incomplete	13 (16.9%)	1 (1.0%)
Wrong answer judged as correct	2 (2.6%)	21 (21.9%)
Wrong answer judged as over-informative	19 (24.7%)	11 (11.5%)
Wrong answer judged as incomplete	9 (11.7%)	30 (31.2%)
Incomplete answer judged as correct	0 (0.0%)	3 (3.1%)
Incomplete answer judged as over-informative	0 (0.0%)	1 (1.0%)
Total	**77**	**96**

Table 10: Manual qualitative classification of LLM judge errors for POLLUX and DeepSeek-R1. The analyzed errors account for 5.0% and 6.2% of all 1,543 benchmark questions, respectively; row percentages are computed within each judge's error set.

In addition to the aggregate agreement metrics, we conducted a manual qualitative analysis of judge errors. The goal of this analysis was to identify recurring failure modes of the LLM-as-a-judge setup and to better understand how these failures may affect the interpretation of the benchmark results.

A linguist manually inspected cases in which the automatic judge verdict did not match the human annotation and assigned each case to an error category. The annotation focused on the direction of the judge error: whether the judge underestimated the answer quality, overestimated it, or made a score-level distinction that did not affect the final binary correctness decision.

Table 10 summarizes the resulting error taxonomy. For `POLLUX`, the analysis includes 77 judge error cases, corresponding to 5.0% of all questions. For `DeepSeek-R1`, the analysis includes 96 judge error cases after filtering out cases that were excluded from the qualitative analysis, corresponding to 6.2% error rate.

The qualitative analysis shows that the two judges have different error profiles. `POLLUX` tends to be more conservative: among binary-relevant errors, 37 cases correspond to underestimation of answer quality, while 21 cases correspond to overestimation. Thus, 63.8% of `POLLUX` binary-relevant errors are underestimations. The most frequent `POLLUX` error types are judging correct answers as incomplete and judging wrong answers as over-informative, each accounting for 19 cases. Another frequent failure mode is treating over-informative answers as incomplete, which occurred in 13 cases.

`DeepSeek-R1` shows the opposite tendency. Among binary-relevant errors, 11 cases correspond to underestimation, whereas 36 cases correspond to overestimation. Thus, 76.6% of `DeepSeek-R1` binary-relevant errors are overestimations. The largest category is wrong answers judged as incomplete, with 30 cases. Although this category does not necessarily change the final binary decision if both wrong and incomplete answers are mapped to incorrect, it indicates that the judge often recognizes that the answer is imperfect but fails to identify that the semantic content is fundamentally wrong. More importantly, `DeepSeek-R1` also frequently assigns positive judgments to incorrect answers: 21 wrong answers were judged as correct and 11 wrong answers were judged as over-informative.

We further grouped reasoning-level failures into broader categories. For both judges, the dominant source of error is incorrect identification of the semantic core of the answer. This accounts for 50 cases for `POLLUX` and 87 cases for `DeepSeek-R1`. In these cases, the judge fails to correctly determine whether the model answer preserves the key meaning of the reference answer. `POLLUX` also shows a relatively larger share of cases where the judge reasoning is mostly plausible but the final verdict is inconsistent with that reasoning. This occurred in 21 cases. In contrast, `DeepSeek-R1` rarely distorts the rubric explicitly, but more often fails at semantic equivalence judgments.

These findings suggest that the judge errors are not random. The Russian judge is biased toward stricter scoring and tends to penalize answers for incompleteness or unsupported extra details. The English judge is more permissive and more often overestimates semantically incorrect answers.

F Detailed Slice-Level Result Analysis

This appendix provides the detailed slice-level analysis underlying the summary in Section 5. The purpose of this analysis is to use the full set of benchmark annotations to obtain more fine-grained insights into method behavior. These annotations capture different properties of memory-oriented question answering, including the semantic operation required by the question, the distribution of supporting evidence across sessions, and the need for temporal grounding.

Each question belongs to three main axes: semantic type, single-session versus multi-session evidence, and temporal versus atemporal setting. We additionally analyze temporal-expression explicitness, language effects, evidence-position effects, and semantic supergroups. Unless stated otherwise, all analyses use LLM-as-Judge accuracy as the primary metric.

Let $y^{\ell}_{m,q} \in \{0,1\}$ denote the binary correctness score of method m on question q in language ℓ. For a method family F, we define the per-question family score as

$$s^{\ell}_{F,q} = \frac{1}{|F|} \sum_{m \in F} y^{\ell}_{m,q}.$$

For a benchmark slice A with question set Q_A, the slice-level family score is

$$\bar{s}^{\ell}_{F,A} = \frac{1}{|Q_A|} \sum_{q \in Q_A} s^{\ell}_{F,q}.$$

Family-level comparisons within a slice are computed as

$$\Delta^{\ell}_{\mathrm{FC-Agent/RAG},A} = \bar{s}^{\ell}_{\mathrm{FC},A} - \bar{s}^{\ell}_{\mathrm{Agent/RAG},A},$$

whereas slice-difficulty contrasts within a family compare two slices A and B as

$$\Delta^{\ell}_{A-B,F} = \bar{s}^{\ell}_{F,A} - \bar{s}^{\ell}_{F,B}.$$

Thus, slices of different sizes are compared through their own mean per-question scores rather than raw counts. Confidence intervals are estimated by bootstrap resampling over questions; p-values are Holm-corrected over planned comparisons.

Slice-level results are reported separately for Agent/RAG and full-context systems, and all analyses are reported independently for Russian and English. We also provide descriptive method-level results to compare individual baselines within and across the two method families. For temporal, temporal-expression, and scope–temporal family aggregates, the Agent/RAG family excludes `mem0g`, whose graph-based setup is atemporal; all other aggregates retain the full family.

All code used to produce slice-level analyses, statistical tests, tables, and figures is open source.

F.1 Full Context vs Agent/RAG as Method Families

Table 11 reports the *non-direct family-level comparison* across the main benchmark slices, without matching the answer model. Full-context scores significantly exceed Agent/RAG scores in all listed slices for both languages. The overall gap is larger in English (+15.11 accuracy points) than in Russian (+11.86 accuracy points). The difference between these two gaps is also significant: $(\Delta_{\mathrm{EN}} - \Delta_{\mathrm{RU}}) = +3.25$ accuracy points, 95% CI $[1.82, 4.72]$. The matched `gpt-4.1-mini` comparison in Table 6 yields smaller overall gaps.

F.2 Single-Session vs Multi-Session Difficulty

Table 12 shows that multi-session questions are consistently harder than single-session questions. The single-to-multi degradation ranges from +20.73 to +22.96 accuracy points across languages and families. This makes session multiplicity the strongest difficulty axis in the benchmark.

Importantly, the multi-session penalty is not specific to retrieval-based systems. Full-context systems also show a large degradation from single-session to multi-session questions. Additional interaction tests show that the degradation difference between full-context and Agent/RAG families is not statistically significant: in Russian, the difference is +2.23 accuracy points with 95% CI $[-0.69, 5.13]$; in English, it is +0.46 accuracy points with 95% CI $[-2.42, 3.42]$.

F.3 Temporal vs Atemporal Difficulty

Table 13 shows that temporal questions are harder on average, but the effect is smaller than the single-to-multi degradation. The atemporal-to-temporal gap ranges from +2.14 to +7.15 accuracy points. The largest temporal penalty appears for English Agent/RAG systems, while the Russian Agent/RAG contrast is not significant after correction.

In English, the Full-context–Agent/RAG gap is +12.43 accuracy points on temporal questions and +10.89 on atemporal questions. In Russian, the corresponding gaps are +4.55 and +8.84, respectively.

F.4 Joint Scope–Temporal Intersections

Table 14 reports the four intersections of the scope and temporal axes. The hardest setting is the multi-session + temporal intersection. The multi-session

Slice	Lang.	Full context	Agent/RAG	Δ	95% CI	Holm p
Overall	RU	67.66	55.80	+11.86	[10.56, 13.17]	<0.001
Overall	EN	69.40	54.29	+15.11	[13.80, 16.42]	<0.001
Single	RU	76.65	63.92	+12.73	[11.12, 14.36]	<0.001
Single	EN	78.21	62.92	+15.29	[13.66, 16.91]	<0.001
Multi	RU	53.69	43.18	+10.51	[8.27, 12.75]	<0.001
Multi	EN	55.70	40.87	+14.83	[12.61, 17.10]	<0.001
Atemporal	RU	69.39	60.55	+8.84	[7.27, 10.39]	<0.001
Atemporal	EN	70.91	60.02	+10.89	[9.32, 12.40]	<0.001
Temporal	RU	62.96	58.41	+4.55	[1.77, 7.32]	0.001
Temporal	EN	65.30	52.87	+12.43	[9.54, 15.28]	<0.001
Single + atemporal	RU	78.61	69.00	+9.62	[7.77, 11.50]	<0.001
Single + atemporal	EN	80.11	70.03	+10.08	[8.18, 11.91]	<0.001
Multi + atemporal	RU	55.49	47.82	+7.67	[4.91, 10.39]	<0.001
Multi + atemporal	EN	57.05	44.93	+12.11	[9.40, 14.80]	<0.001
Single + temporal	RU	71.54	67.20	+4.34	[0.83, 7.93]	0.027
Single + temporal	EN	73.29	60.38	+12.91	[9.28, 16.53]	<0.001
Multi + temporal	RU	48.42	43.51	+4.92	[0.56, 9.30]	0.029
Multi + temporal	EN	51.76	40.13	+11.63	[6.75, 16.57]	<0.001

Table 11: Family-level comparison between Full context and Agent/RAG across the main benchmark slices. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ = Full context − Agent/RAG. Agent/RAG temporal and scope–temporal aggregates exclude `mem0g`.

Family	Lang.	Single	Multi	Δ	95% CI	Holm p
Agent/RAG	RU	63.92	43.18	+20.73	[17.71, 23.83]	<0.001
Agent/RAG	EN	62.92	40.87	+22.06	[19.02, 25.14]	<0.001
Full context	RU	76.65	53.69	+22.96	[20.07, 25.88]	<0.001
Full context	EN	78.21	55.70	+22.51	[19.57, 25.50]	<0.001

Table 12: Single-session versus multi-session difficulty within method families. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ = Single − Multi.

penalty is significant within both atemporal and temporal questions. Within single-session questions, the temporal penalty is significant for English Agent/RAG and both full-context languages, but not for Russian Agent/RAG. Within multi-session questions, it is not significant for Agent/RAG in either language or for English full context after correction.

F.5 Temporal-Expression Explicitness

Table 16 first reports the Full context versus Agent/RAG comparison (in a non-answerer-fixed setting) within each temporal expression tag. The gap is significant for explicit expressions in both languages and for all English tags; the Russian implicit and no-expression gaps are small and not significant after correction.

Table 17 shows that implicit temporal expressions are much harder than explicit temporal expressions. This is one of the strongest slice-level findings. For full-context systems, the explicit-to-implicit gap is approximately 30.02 accuracy points in Russian and 29.89 accuracy points in English. Thus, simply providing the full dialogue does not eliminate the difficulty of implicit temporal grounding.

Questions with no explicit temporal expression occupy an intermediate regime: they are significantly easier than questions with implicit temporal evidence, but for full-context systems they are significantly harder than questions with explicit temporal evidence.

F.6 Language Effects

Table 18 reports English–Russian differences. *These comparisons should be interpreted cautiously because Russian and English runs use language-specific judges.* At the family level, language effects are statistically detectable but small: Agent/RAG systems perform slightly better in Russian overall, whereas full-context systems perform slightly better in English overall. The clearest language-conditioned difference for Agent/RAG appears on temporal questions and implicit/no-expression temporal tags, where Russian scores are higher. For full-context systems, the largest language effect appears in the Reasoning semantic supergroup.

F.7 Semantic Supergroups

We collapse the first category axis into three semantic "supergroups": Extraction, Reasoning, and

Family	Lang.	Atemporal	Temporal	Δ	95% CI	Holm p
Agent/RAG	RU	60.55	58.41	+2.14	[-1.55, 5.74]	0.259
Agent/RAG	EN	60.02	52.87	+7.15	[3.54, 10.70]	<0.001
Full context	RU	69.39	62.96	+6.43	[3.10, 9.67]	<0.001
Full context	EN	70.91	65.30	+5.61	[2.23, 8.93]	0.004

Table 13: Atemporal versus temporal difficulty within method families. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ = Atemporal − Temporal. Agent/RAG aggregates exclude `mem0g`.

Family	Lang.	Single + Atemporal	Multi + Atemporal	Single + Temporal	Multi + Temporal
Agent/RAG	RU	69.00	47.82	67.20	43.51
Agent/RAG	EN	70.03	44.93	60.38	40.13
Full context	RU	78.61	55.49	71.54	48.42
Full context	EN	80.11	57.05	73.29	51.76

Table 14: Family-level accuracy on the 4 scope–temporal intersections. Scores are family-level means of binary LLM-as-judge correctness and are reported on a 0–100 accuracy-point scale. The table shows how session scope and temporal reasoning jointly affect performance within each method family and language. Agent/RAG aggregates exclude `mem0g`.

Abstention (Table 2).

Table 19 shows that Abstention is comparatively saturated: Agent/RAG systems and full-context systems perform similarly (in a setting without a matched answer model), and the gap is not statistically significant in either language. In contrast, full-context systems significantly outperform Agent/RAG systems on both Reasoning and Extraction. Reasoning is the hardest supergroup for both families.

Table 20 reports the within-family semantic-supergroup difficulty contrasts. Extraction is consistently easier than Reasoning. Abstention is easier than both Extraction and Reasoning, especially relative to Reasoning. These contrasts show that the semantic grouping captures a robust difficulty gradient rather than only a difference between method families.

F.8 Reasoning vs Temporal Reasoning

As a more focused reasoning analysis, we compare non-temporal Reasoning categories against Temporal Reasoning categories from Fig.2. Table 21 reports the planned contrasts.

The observed differences are small and not statistically significant after Holm correction. Moreover, the direction of the difference is not consistent across languages and method families: Temporal Reasoning is slightly higher for Agent/RAG in Russian, whereas general Reasoning is slightly higher in the other three settings.

Although we do not observe a significant aggregate effect in this comparison, the two groups differ conceptually and may still reveal model-specific patterns for individual model/methods.

F.9 Semantic-Type Heatmaps

Figures 15 and 16 show method-level heatmaps by semantic type. These figures are intended as descriptive visualizations rather than statistical comparisons, since semantic categories vary in sample size.

F.10 Lost-in-the-Middle Proxy

We also test whether performance drops when the evidence is located in the middle of the dialogue. Since exact token positions in the final prompt are not used here, this is a session-level proxy (thus, it cannot adjudicate token-level lost-in-the-middle effects): questions with evidence near the beginning or end are grouped as edge, and compared against questions with evidence in the middle.

Table 22 shows no statistically significant edge-versus-middle effect for either method family or language. Thus, we do not detect a lost-in-the-middle pattern under this proxy.

Because evidence-position effects can be confounded by multi-session questions, we also repeat the same edge-versus-middle analysis on single-session questions only. Table 23 shows the same conclusion: no statistically significant lost-in-the-middle effect is detected after restricting to single-session questions.

Contrast	Family / Lang.	A	B	Δ	95% CI	Holm p
Single+Atemporal − Multi+Atemporal	Agent/RAG RU	69.00	47.82	+21.17	[17.18, 25.03]	<0.001
Single+Atemporal − Multi+Atemporal	Agent/RAG EN	70.03	44.93	+25.10	[21.13, 28.97]	<0.001
Single+Atemporal − Multi+Atemporal	Full context RU	78.61	55.49	+23.12	[19.62, 26.42]	<0.001
Single+Atemporal − Multi+Atemporal	Full context EN	80.11	57.05	+23.06	[19.53, 26.54]	<0.001
Single+Temporal − Multi+Temporal	Agent/RAG RU	67.20	43.51	+23.70	[17.67, 29.77]	<0.001
Single+Temporal − Multi+Temporal	Agent/RAG EN	60.38	40.13	+20.25	[14.10, 26.33]	<0.001
Single+Temporal − Multi+Temporal	Full context RU	71.54	48.42	+23.12	[17.50, 28.64]	<0.001
Single+Temporal − Multi+Temporal	Full context EN	73.29	51.76	+21.53	[15.69, 27.33]	<0.001
Single+Atemporal − Single+Temporal	Agent/RAG RU	69.00	67.20	+1.79	[-2.30, 6.02]	0.406
Single+Atemporal − Single+Temporal	Agent/RAG EN	70.03	60.38	+9.65	[5.33, 14.05]	<0.001
Single+Atemporal − Single+Temporal	Full context RU	78.61	71.54	+7.08	[3.32, 10.67]	<0.001
Single+Atemporal − Single+Temporal	Full context EN	80.11	73.29	+6.82	[3.29, 10.47]	<0.001
Multi+Atemporal − Multi+Temporal	Agent/RAG RU	47.82	43.51	+4.32	[-1.58, 10.15]	0.207
Multi+Atemporal − Multi+Temporal	Agent/RAG EN	44.93	40.13	+4.80	[-0.96, 10.57]	0.207
Multi+Atemporal − Multi+Temporal	Full context RU	55.49	48.42	+7.07	[1.61, 12.38]	0.045
Multi+Atemporal − Multi+Temporal	Full context EN	57.05	51.76	+5.29	[-0.46, 10.84]	0.206

Table 15: Within-family contrasts over the scope–temporal intersections. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ computed as the first slice minus the second slice in each contrast. Agent/RAG aggregates exclude `mem0g`.

Temporal-expression tag	Lang.	Full context	Agent/RAG	Δ	95% CI	Holm p
Explicit	RU	76.19	63.33	+12.86	[8.07, 17.79]	<0.001
Explicit	EN	76.89	60.20	+16.69	[11.93, 21.60]	<0.001
Implicit	RU	46.17	44.35	+1.82	[-3.60, 7.16]	0.692
Implicit	EN	47.00	38.55	+8.45	[1.41, 15.40]	0.017
No temporal expression	RU	62.18	60.33	+1.85	[-1.89, 5.70]	0.692
No temporal expression	EN	65.63	53.85	+11.78	[7.88, 15.71]	<0.001

Table 16: Family-level comparison between Full context and Agent/RAG by temporal expression tags. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ = Full context − Agent/RAG. Agent/RAG aggregates exclude `mem0g`.

F.11 Best-Observed Methods

For practical interpretation, we also report the best observed method in each family. This is an upper-bound comparison and is not used as the main statistical evidence for family-level claims.

Across most full-context slices, the strongest observed model is GPT-5.4. For Agent/RAG systems, the strongest Russian results are typically obtained by simple RAG, cortex, or memOS depending on the slice; in English, memOS is the strongest Agent/RAG method across most major slices. These comparisons characterize the strongest observed upper bounds, not a controlled effect of the memory-access mechanism; the matched-answerer results in Table 6 show that individual Agent/RAG systems can exceed the `gpt-4.1-mini` full-context baseline.

G Case study: Claude Sonnet 4.6

In this section, we demonstrate how RUMBA can be used to profile a single selected model rather than for benchmarking different methods. For this case study, we choose `anthropic/claude-sonnet-4.6`, the second-best system in the full-context setting in both language versions of our evaluation, reaching 78.61 accuracy points in Russian and 81.98 accuracy points in English. Using the same diagnostic axes and statistical procedure as in the preceding appendix analyses, we examine where this model is robust, where it degrades, and which benchmark dimensions explain its main failure modes.

Table 24 reports descriptive slice-level scores. Claude Sonnet 4.6 is strongest on Abstention and Extraction-oriented questions, while its weaker regions are multi-session questions, implicit temporal expressions, and Reasoning-oriented questions. This profile is broadly consistent across languages, although English is generally stronger than Russian.

Table 25 summarizes the statistically significant contrasts. All deltas are reported in accuracy points on a 0–100 scale. We omit non-significant contrasts from the table to keep the case study focused on the model's main diagnostic failure modes.

The clearest degradation is caused by session scope. As shown in Table 24, accuracy drops from 83.49 to 71.03 in Russian and from 88.60 to 71.69 in English when moving from single-session to multi-session

Contrast	Family / Lang.	A	B	Δ	95% CI	Holm p
Explicit − Implicit	Agent/RAG RU	63.33	44.35	+18.99	[9.68, 28.61]	<0.001
Explicit − Implicit	Agent/RAG EN	60.20	38.55	+21.65	[11.82, 31.34]	<0.001
Explicit − Implicit	Full context RU	76.19	46.17	+30.02	[21.90, 37.77]	<0.001
Explicit − Implicit	Full context EN	76.89	47.00	+29.89	[21.61, 38.35]	<0.001
No expression − Explicit	Agent/RAG RU	60.33	63.33	-3.01	[-9.69, 3.82]	0.387
No expression − Explicit	Agent/RAG EN	53.85	60.20	-6.34	[-13.05, 0.61]	0.153
No expression − Explicit	Full context RU	62.18	76.19	-14.01	[-19.94, -7.71]	<0.001
No expression − Explicit	Full context EN	65.63	76.89	-11.26	[-17.02, -5.20]	<0.001
No expression − Implicit	Agent/RAG RU	60.33	44.35	+15.98	[7.33, 24.95]	<0.001
No expression − Implicit	Agent/RAG EN	53.85	38.55	+15.30	[6.73, 24.30]	<0.001
No expression − Implicit	Full context RU	62.18	46.17	+16.01	[8.75, 23.20]	<0.001
No expression − Implicit	Full context EN	65.63	47.00	+18.63	[11.05, 26.59]	<0.001

Table 17: Temporal-expression explicitness analysis within method families. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ computed as the first tag minus the second tag in each contrast. Agent/RAG aggregates exclude `mem0g`.

Slice	Family	English	Russian	EN–RU	95% CI	Holm p
Overall	Agent/RAG	54.29	55.80	-1.51	[-2.78, -0.24]	0.019
Overall	Full context	69.40	67.66	+1.74	[0.72, 2.74]	0.001
Single	Agent/RAG	62.92	63.92	-0.99	[-2.59, 0.57]	0.217
Single	Full context	78.21	76.65	+1.57	[0.27, 2.85]	0.042
Multi	Agent/RAG	40.87	43.18	-2.32	[-4.42, -0.19]	0.040
Multi	Full context	55.70	53.69	+2.01	[0.35, 3.62]	0.040
Atemporal	Agent/RAG	60.02	60.55	-0.53	[-2.16, 1.06]	0.524
Atemporal	Full context	70.91	69.39	+1.52	[0.35, 2.66]	0.025
Temporal	Agent/RAG	52.87	58.41	-5.54	[-8.05, -3.08]	<0.001
Temporal	Full context	65.30	62.96	+2.34	[0.31, 4.41]	0.023
Implicit time	Agent/RAG	38.55	44.35	-5.80	[-10.72, -0.87]	0.049
Implicit time	Full context	47.00	46.17	+0.83	[-3.73, 5.38]	0.762
No temporal expression	Agent/RAG	53.85	60.33	-6.48	[-9.75, -3.20]	<0.001
No temporal expression	Full context	65.63	62.18	+3.45	[0.88, 6.09]	0.010
Extraction	Agent/RAG	53.82	56.12	-2.30	[-3.83, -0.77]	0.005
Extraction	Full context	69.85	69.20	+0.65	[-0.62, 1.89]	0.325
Reasoning	Agent/RAG	41.67	41.88	-0.22	[-3.12, 2.63]	0.884
Reasoning	Full context	60.51	55.02	+5.48	[3.18, 7.83]	<0.001
Abstention	Agent/RAG	83.01	81.60	+1.41	[-1.62, 4.44]	0.382
Abstention	Full context	84.14	82.28	+1.86	[-0.56, 4.36]	0.294

Table 18: Language comparison. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with $\Delta = \text{English} - \text{Russian}$. Agent/RAG temporal and temporal-expression aggregates exclude `mem0g`. Note that Russian and English evaluations use different LLM judges.

questions. The corresponding contrasts in Table 25 are significant in both languages, with gaps of 12.47 and 16.92 accuracy points. This degradation also remains significant inside both atemporal and temporal subsets. The hardest scope–temporal intersection is multi-session temporal questioning, where the model reaches 66.88 in Russian and 66.23 in English. This suggests that the main structural weakness is not temporal grounding alone, but the combination of temporal grounding with evidence distributed across multiple sessions.

Temporal questions as a broad category are not the strongest explanation of the model's failures. The atemporal–temporal contrast is not significant after Holm correction in either language. However, the temporal-expression analysis reveals a more specific failure mode. Claude handles explicit temporal expressions well, with accuracy of 85.29 in Russian and 84.31 in English, but drops sharply on implicit temporal expressions, reaching only 57.97 and 60.87. The explicit–implicit gaps are large and significant in both languages: 27.32 accuracy points in Russian and 23.44 in English. Implicit temporal expressions are also significantly harder than questions with no temporal expression.

The semantic profile shows another major weakness. Claude is strong on Abstention, especially in Russian, where it reaches 90.26 accuracy points, and it is also relatively strong on Extraction. Reasoning is substantially weaker: 65.16 in Russian and 76.77 in English. The Extraction–Reasoning contrast is significant in both languages, while Reasoning–Abstention is even larger, especially in Russian, where the gap is -25.10 accuracy points. This indicates that the model is reliable when it must retrieve or reject unknown user information,

Semantic supergroup	Lang.	Full context	Agent/RAG	Δ	95% CI	Holm p
Abstention	RU	82.28	81.60	+0.68	[-2.69, 4.13]	0.694
Abstention	EN	84.14	83.01	+1.13	[-1.73, 4.00]	0.455
Reasoning	RU	55.02	41.88	+13.14	[10.21, 16.12]	<0.001
Reasoning	EN	60.51	41.67	+18.84	[15.71, 22.07]	<0.001
Extraction	RU	69.20	56.12	+13.09	[11.45, 14.70]	<0.001
Extraction	EN	69.85	53.82	+16.04	[14.43, 17.60]	<0.001

Table 19: Family-level comparison between Full context and Agent/RAG by semantic supergroup from Figure 2. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ = Full context − Agent/RAG.

Contrast	Family / Lang.	A	B	Δ	95% CI	Holm p
Extraction − Reasoning	Agent/RAG RU	56.12	41.88	+14.24	[10.26, 18.11]	<0.001
Extraction − Reasoning	Agent/RAG EN	53.82	41.67	+12.15	[8.32, 16.02]	<0.001
Extraction − Reasoning	Full context RU	69.20	55.02	+14.18	[10.22, 18.19]	<0.001
Extraction − Reasoning	Full context EN	69.85	60.51	+9.35	[5.48, 13.33]	<0.001
Extraction − Abstention	Agent/RAG RU	56.12	81.60	-25.48	[-29.65, -21.31]	<0.001
Extraction − Abstention	Agent/RAG EN	53.82	83.01	-29.19	[-33.52, -24.73]	<0.001
Extraction − Abstention	Full context RU	69.20	82.28	-13.08	[-17.17, -9.01]	<0.001
Extraction − Abstention	Full context EN	69.85	84.14	-14.28	[-18.47, -9.95]	<0.001
Reasoning − Abstention	Agent/RAG RU	41.88	81.60	-39.72	[-44.85, -34.48]	<0.001
Reasoning − Abstention	Agent/RAG EN	41.67	83.01	-41.34	[-46.58, -36.15]	<0.001
Reasoning − Abstention	Full context RU	55.02	82.28	-27.26	[-32.25, -22.08]	<0.001
Reasoning − Abstention	Full context EN	60.51	84.14	-23.63	[-28.71, -18.46]	<0.001

Table 20: Within-family difficulty contrasts between semantic supergroups from Figure 2. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ computed as the first supergroup minus the second supergroup in each contrast.

but less reliable when the answer requires combining, comparing, or transforming it.

Temporal Reasoning is not the uniquely problematic group for Claude Sonnet 4.6. In Russian, Temporal Reasoning is numerically higher than general Reasoning, 69.57 versus 60.40 accuracy points. In English, the direction reverses: General Reasoning reaches 78.52, while Temporal Reasoning reaches 75.16. The within-language Reasoning–Temporal Reasoning contrast is not significant after correction in either language. However, the English–Russian gap is significant for general Reasoning, with English outperforming Russian by 18.12 accuracy points, while the corresponding language gap for Temporal Reasoning is not significant. This suggests that Claude's Russian weakness is concentrated more in general memory reasoning operations than in Temporal Reasoning specifically.

The language comparison also reveals a targeted rather than uniform gap. English is significantly better overall by 3.37 accuracy points, but the largest language differences occur in Reasoning-heavy slices. The English–Russian gap is 11.61 accuracy points for the Reasoning supergroup and 18.12 accuracy points for General Reasoning. By contrast, multi-session questions, temporal questions as a whole, implicit temporal expressions, Abstention, and Temporal Reasoning do not show significant English–Russian differences. Therefore, Claude's Russian weakness is concentrated primarily in General Reasoning rather than in memory retrieval or temporal grounding overall.

Finally, the evidence-position analysis does not support a clear lost-in-the-middle explanation for this model. The edge–middle contrast is not significant in either language, including when restricting the analysis to single-session questions. For all questions, edge–middle differences are -0.30 accuracy points in Russian and 3.54 in English; for single-session questions only, they are -2.78 and 0.83 accuracy points, respectively. Consequently, the dominant failure modes are better explained by multi-session integration, implicit temporal inference, and Reasoning operations than by the absolute position of the supporting evidence in the dialogue.

Contrast	Family / Lang.	A	B	Δ	95% CI	Holm p
Reasoning − TR	Agent/RAG RU	39.49	44.10	-4.61	[-11.54, 2.35]	0.792
Reasoning − TR	Agent/RAG EN	42.62	40.79	+1.83	[-4.86, 8.68]	1.000
Reasoning − TR	Full context RU	55.99	54.13	+1.87	[-5.16, 8.95]	1.000
Reasoning − TR	Full context EN	62.42	58.74	+3.68	[-2.94, 10.18]	0.868

Table 21: General Reasoning versus Temporal Reasoning (TR) categories defined in Figure 2. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ computed as General Reasoning minus Temporal Reasoning.

Family	Lang.	Edge	Middle	Edge–Middle	95% CI	Holm p
Agent/RAG	RU	55.62	56.13	-0.51	[-3.68, 2.66]	0.986
Agent/RAG	EN	55.37	52.36	+3.01	[-0.20, 6.26]	0.270
Full context	RU	68.08	66.90	+1.18	[-1.96, 4.30]	0.986
Full context	EN	69.97	68.37	+1.60	[-1.60, 4.83]	0.986

Table 22: Lost-in-the-middle proxy analysis by evidence position in the dialogue. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ computed as Edge minus Middle. Edge combines questions whose evidence is located near the beginning or end of the dialogue.

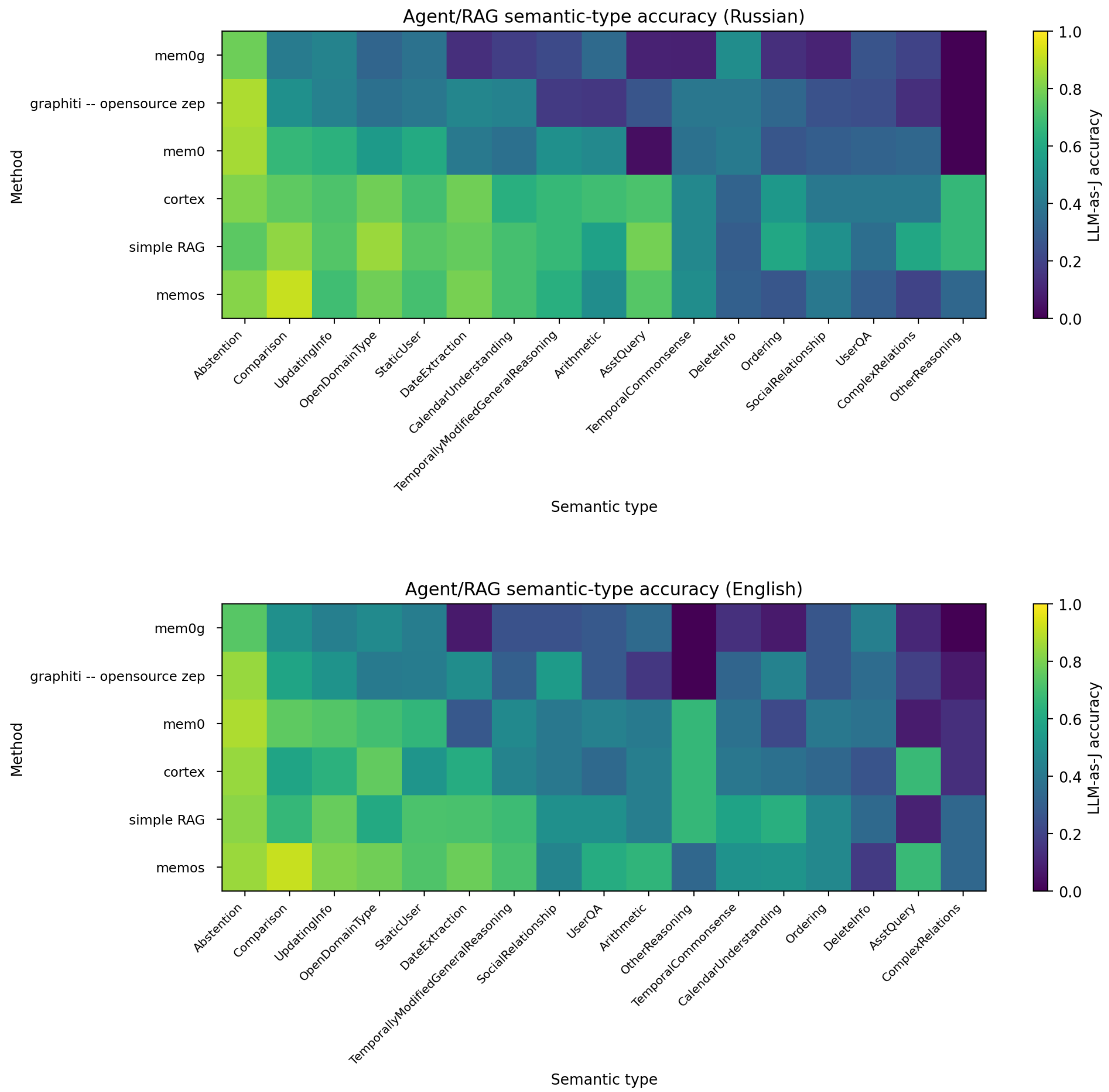
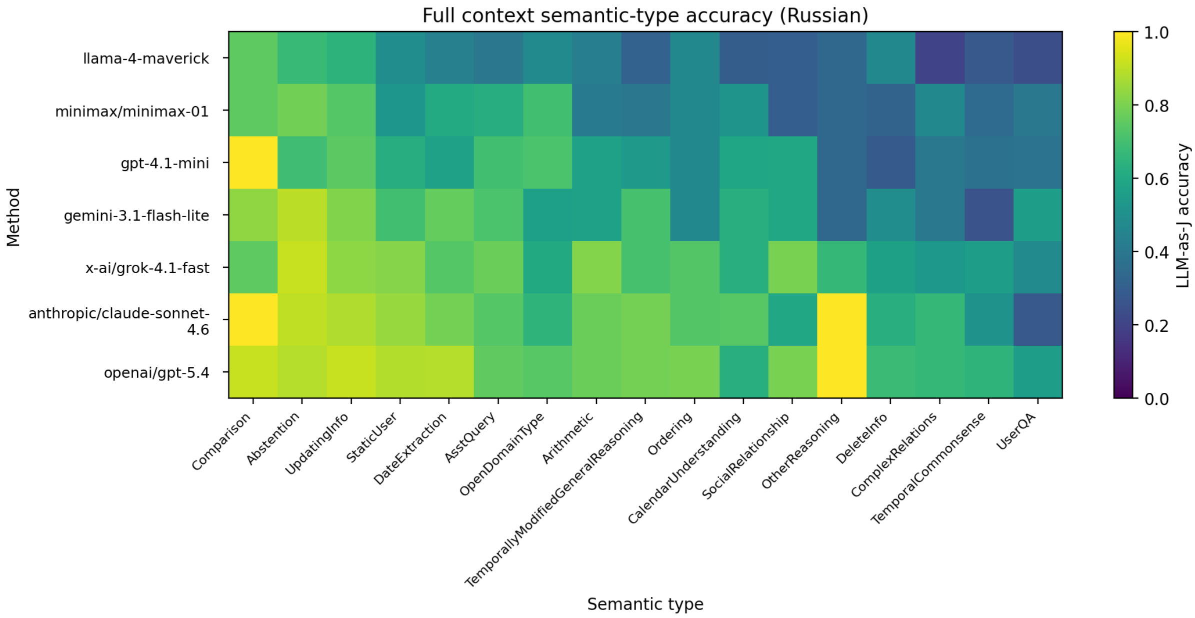


Figure 15: Agent/RAG heatmaps by semantic type.



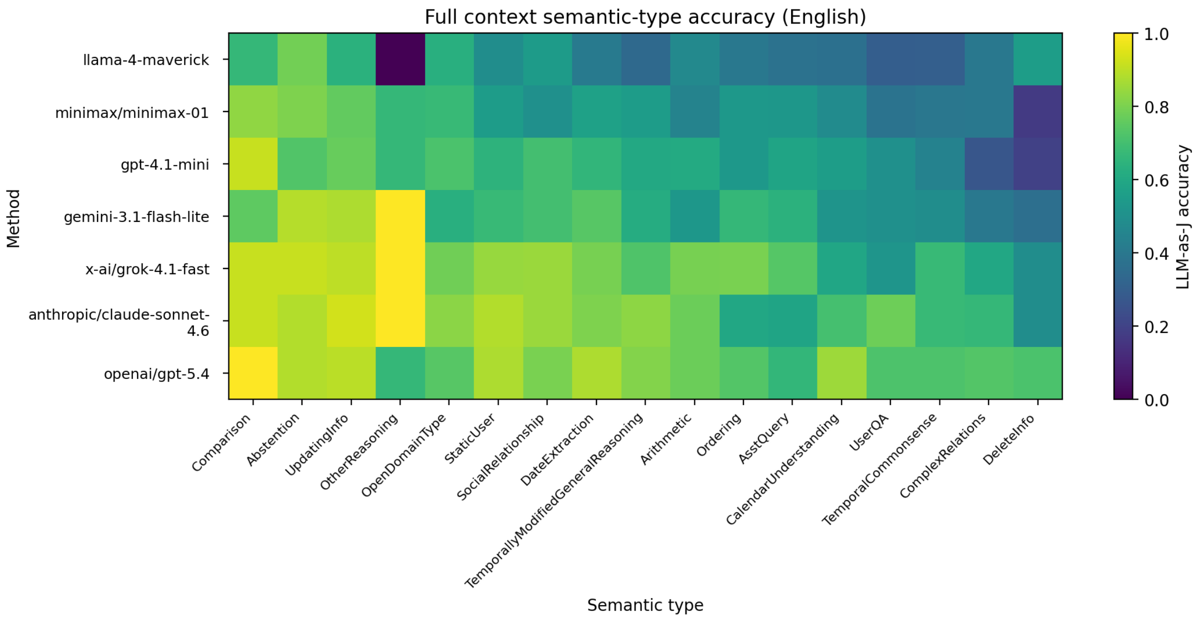


Figure 16: Full-context heatmaps by semantic type.

Family	**Lang.**	**Edge**	**Middle**	**Edge–Middle**	**95% CI**	**Holm p**
Agent/RAG	RU	63.87	64.01	-0.14	[-4.04, 3.64]	1.000
Agent/RAG	EN	64.11	60.41	+3.70	[-0.28, 7.72]	0.289
Full context	RU	76.11	77.79	-1.68	[-5.13, 1.90]	1.000
Full context	EN	77.83	79.02	-1.19	[-4.73, 2.47]	1.000

Table 23: Lost-in-the-middle proxy analysis restricted to single-session questions. Scores are family-level means of binary LLM-as-judge correctness; scores and differences are reported on a 0–100 accuracy-point scale, with Δ computed as Edge minus Middle.

Axis	Slice	n	Russian	English
Overall	All questions	1543	78.61	81.98
Scope	Single-session	939	83.49	88.60
	Multi-session	604	71.03	71.69
Temporal grounding	Atemporal	1128	79.79	83.24
	Temporal	415	75.42	78.55
Scope $\times$ time	Single + atemporal	678	84.66	89.68
	Multi + atemporal	450	72.44	73.56
	Single + temporal	261	80.46	85.82
	Multi + temporal	154	66.88	66.23
Temporal expression	Explicit	102	85.29	84.31
	Implicit	69	57.97	60.87
	No temporal expression	244	76.23	81.15
Semantic supergroup	Extraction	1079	80.82	82.58
	Reasoning	310	65.16	76.77
	Abstention	154	90.26	88.31
Reasoning supergroup	General Reasoning	149	60.40	78.52
	Temporal Reasoning	161	69.57	75.16

Table 24: Single-model diagnostic profile for Claude Sonnet 4.6 across benchmark axes and slices. Values are LLM-as-judge accuracy points on a 0–100 scale; n denotes the number of questions in each slice.

Block	Contrast	Lang.	Δ	95% CI	Holm p
Scope	Single $-$ multi	RU	12.47	[8.21, 16.77]	< 0.001
	Single $-$ multi	EN	16.92	[12.82, 21.02]	< 0.001
Scope $\times$ time	Single+atemporal $-$ multi+atemporal	RU	12.22	[7.40, 17.02]	< 0.001
	Single+atemporal $-$ multi+atemporal	EN	16.12	[11.47, 20.71]	< 0.001
	Single+temporal $-$ multi+temporal	RU	13.58	[4.93, 22.39]	0.003
	Single+temporal $-$ multi+temporal	EN	19.59	[11.27, 28.12]	< 0.001
Temporal expression	Explicit $-$ implicit	RU	27.32	[13.68, 40.79]	< 0.001
	Explicit $-$ implicit	EN	23.44	[9.85, 36.62]	0.001
	No expression $-$ implicit	RU	18.26	[5.43, 31.30]	0.005
	No expression $-$ implicit	EN	20.28	[7.67, 32.91]	0.003
Semantic supergroup	Extraction $-$ Reasoning	RU	15.65	[9.84, 21.62]	< 0.001
	Extraction $-$ Reasoning	EN	5.80	[0.55, 11.06]	0.030
	Extraction $-$ Abstention	RU	-9.44	[-14.55, -3.97]	0.003
	Reasoning $-$ Abstention	RU	-25.10	[-32.20, -17.66]	< 0.001
	Reasoning $-$ Abstention	EN	-11.54	[-18.33, -4.42]	0.002
English $-$ Russian	Overall	–	3.37	[1.23, 5.44]	< 0.001
	Single-session	–	5.11	[2.77, 7.56]	< 0.001
	Atemporal	–	3.46	[1.15, 5.76]	0.006
	Single+atemporal	–	5.01	[2.21, 7.82]	< 0.001
	Single+temporal	–	5.36	[0.38, 10.34]	0.049
	Reasoning supergroup	–	11.61	[6.77, 16.45]	< 0.001
	General Reasoning	–	18.12	[10.07, 25.50]	< 0.001

Table 25: Significant single-model diagnostic contrasts for Claude Sonnet 4.6 across benchmark axes. Differences are reported as LLM-as-judge accuracy points on a 0–100 scale; Δ is computed as the first slice minus the second slice in each contrast, except for the language block where Δ = English $-$ Russian.

Table 26: Question taxonomy: detailed description of question types and subtypes according to the semantic axis, with illustrative examples.

Category&Type	Type Description	Example
NON-TEMPORAL		
INFORMATION EXTRACTION		
StaticUser	**(a)** The fact to which the question refers remains unchanged across all sessions, both qualitatively and quantitatively. **(b)** The facts to which the question refers are complementary and do not constitute a qualitative or quantitative change of the same fact across sessions.	**User:** Our dog's name is Losyash. **Question:** What's my dog's name? **Answer:** Losyash.
UpdatingInfo	The fact to which the question refers undergoes qualitative or quantitative changes within one or more sessions.	**Condition:** Pick the most up-to-date answer within one or across two or more sessions. **User:** I love roses. [...] **User:** I really love cacti now, I'm tired of roses. [...] **User:** I don't like cacti anymore, lilies are my favorite now. **Question:** What are my favorite plants? **Answer:** Lilies.
DeleteInfo	The fact to which the question refers is deleted by the user within one or more sessions.	**Condition:** Pick the most up-to-date answer within one or across two or more sessions. **AnswerStandard:** The correct answer is "No such information" (or similar). **User:** I don't want to be a poor peasant woman, I want to be a noble lady. ... **User:** Delete the information about what I want to be. **Question:** What do I want to be? **Answer:** No such information.
AsstQuery	The fact to which the question refers is contained in the assistant's response. In this case, the response constitutes advice, a recommendation, or an answer to an instructional request. Important: the fact does not concern the assistant's persona and does not disclose attributes such as age, gender, personality, name, etc.	**User:** Who should I read from the 20th century? **Assistant:** [List]... Bulgakov, Pasternak, Dovlatov, Orwell, Joyce... **Question:** Which Russian 20th-century writers did you recommend? **Answer:** Bulgakov, Pasternak, Dovlatov.

Continuation

Category&Type	Type Description	Example
OpenDomain Type	The fact to which the question refers pertains to widely known entities (events—past or recent—objects, public figures, etc.) and is characterized by the fact that a correct or approximate answer can be obtained from publicly available sources without consulting the session history.	**User:** In Attack on Titan, my favorite pairing is Jean and Mikasa. **Question:** What's my favorite Attack on Titan pairing? **Answer:** Jean and Mikasa.
REASONING		
Social Relationship	The fact to which the question refers reveals the nature of the relationship between individuals. Important: the social role is not explicitly stated in the dialogue and must be inferred through reasoning.	**User:** I don't have many relatives, my mom only has one sister, Lera. **Question:** Who is Lera to me? **Answer:** Your aunt.
Ordering	The entities to which the question refers share a common attribute. To answer the question, these entities (objects, features, etc.) must be arranged in a specific order according to a given parameter.	**User:** My sisters' names are Ira, Vasilisa, and Alisa. **Question:** List my sisters in alphabetical order. **Answer:** Alisa, Ira, Vasilisa.
Arithmetic	The fact to which the question refers requires arithmetic computation: (a) sum — calculation of the total; (b) less / more — calculation of the difference; (c) average — calculation of the mean value; (d) other — other mathematical operations.	AnswerStandard: The answer should include a number. **User:** I have a guitar, a domra, and my grandma's piano at home. But I only play the guitar and domra. **Question:** How many musical instruments do I have? **Answer:** Three.
Comparison	The fact to which the question refers requires qualitative or quantitative comparison of two or more entities. The question reflects an understanding of comparative, superlative, and equivalence relations between entities: (a) comparative — direct comparison (A is more X than B); (b) superlative — identification of the most X within a set; (c) equivalence — A is equal to B.	**User:** We have two cats: Musya (5 years old) and Kusya (5 months old). **Question:** Which pet is older? **Answer:** Musya.
UserQA	The fact to which the question refers requires an understanding not only of the information about the user contained in the dialogue but also of specific or common world knowledge.	**User:** I professionally study German folklore. **Question:** Am I a Germanist? **Answer:** Yes.
OtherReasoning Other categories of reasoning tasks.		
Causal Reasoning	The fact to which the question refers requires an understanding of causal relationships: why / due to what / what was the cause / what was the effect. Important: both the cause and the effect are present in the text across different sessions and must be correlated. Condition: To answer a question about the user, it is necessary to integrate information from multiple sessions.	**User:** I got into a car accident, the car is wrecked. [...] **User:** I'm in the hospital, good thing the airbag worked. **Question:** Why was I in the hospital? **Answer:** Because you got into a car accident.
MultiStep	The fact to which the question refers requires multi-step reasoning.	**User:** I bought my first houseplants — four violets. [...] **User:** One violet died, Lena gave me her ficus (she had two), and my grandma gave me one big aloe and one small one. [...] **User:** My grandma gave Lena two snake plants because she only had ficuses. **Question:** How many more plants do I have than Lena? **Answer:** By 3.
TEMPORAL		
INFORMATION EXTRACTION		
DateExtraction Types of questions involving the extraction of an event date and/or questions containing a specific date.		

Continuation

Category&Type	Type Description	Example
ExplicitTE Condition Question	**(a)** The user's utterance explicitly contains a date, month, year, or time interval related to an event. **(b)** The question either includes a specific date (day, month, year) and/or refers to such a date. The corresponding subsubtypes: *date-question, month/year-question, span-question, date-answer, month/year-answer, span-answer*	[16.09.2025] **User:** I worked at a swimming pool on July 15. [20.12.2025] **Question:** Where did I work on July 15? [20.12.2025] **Answer:** At a swimming pool.
NonTE Condition Question	**(a)** The user's utterance does not explicitly mention the time of the event and contains no temporal reference. **(b)** The question either includes a specific date (day, month, year) and/or refers to such a date. The corresponding subsubtypes: *date-question, month/year-question, span-question, date-answer, month/year-answer, span-answer*	[01.03.2025] **User:** Come up with a rap nickname for me. [01.03.2025] **Assistant:** [List]... Vupsen... [06.06.2025] **Question:** Do you remember what nickname you gave me in March? **Answer:** Vupsen.
REASONING		
CalendarUnderstanding Types of questions in which it is necessary to correctly understand and interpret calendrical information: terminology and structure, as well as the units used in calendars.		
ImplicitTE Condition Question	**(a)** The user's utterance contains an implicit temporal reference (*yesterday, today, this month, recently, the other day,* etc.). **(b)** A question aimed at identifying and extracting the time of an event. The corresponding subsubtypes: *date-question, month/year-question, span-question, date-answer, month/year-answer, span-answer*	[23.11.2025] **User:** Yesterday I had a singing lesson, today I have a domra lesson. [24.11.2025] **Question:** What lesson did I have on November 23? **Answer:** Domra.
RelativeTime Understanding	A question aimed at understanding relative time, expressed in relation to the moment of speaking (i.e., when the question is asked).	[23.11.2025] **User:** I'm baking a charlotte, but it fell apart. [24.11.2025] **Question:** What did I bake yesterday? **Answer**: Charlotte.
Weekday Understanding	A question concerning the correspondence between a date and a day of the week.	[24.11.2025] **User:** I baked a charlotte on November 20. [25.11.2025] **Question:** What did I bake on Thursday? **Answer:** Charlotte
DateQA	A question concerning calendar periods, production calendars, and the properties of the calendar as a system.	[23.11.2025] **User:** I baked a charlotte on November 13, and today I'm making muffins. [05.12.2025] **Question:** What did I bake in the second half of November? **Answer:** Muffins.
TemporalCommonsense A class of questions that fundamentally requires an understanding of the temporal structure of the world.		
(Event)Ordering	**(a)** A question concerning the sequence of events: what occurred before/after/between, as well as ordinal position: first/second/last. **(b)** A question involving entities sharing a common attribute, based on an understanding of event timing and/or the time of questioning. To answer, these entities must be ordered according to a specified parameter.	[21.05.2025] **User:** I have graduation coming up, and my thesis defense already happened. [...] [11.09.2025] **User:** I started working as a waitress before my thesis defense. [11.12.2025] **Question:** Did I start working as a waitress before graduation? **Answer:** Yes.
EventFrequency	A question concerning event frequency: how often does an event occur? The minimum unit of measurement is one day.	[06.04.2025] **User:** Cucumber for breakfast. [...] [17.04.2025] **User:** Salad with cucumbers. [...] [25.04.2025] **User:** Smashed cucumbers for dinner. [01.05.2025] **Question:** How often do I eat cucumbers? **Answer:** 3 times a month.
EventDuration	A question concerning event duration: how long did X last? how long before did Y occur? how many days after did A take place? The minimum unit of measurement is one day.	[20.10.2025] **User:** I started painting the walls. [...] [25.10.2025] **User:** I got divorced. [30.12.2025] **Question:** How many days before the divorce did I start painting? **Answer:** 5 days.
TemporallyModifiedGeneralReasoning Reasoning questions complicated by temporal constraints.		

Continuation

Category&Type	Type Description	Example
Arithmetic	The fact to which the question refers requires arithmetic computation based on an understanding of event timing and/or the time of questioning.	[11.09.2025] **User:** I weigh 65 kg. [11.10.2025] **Question:** How much do I weigh now if last month I weighed 5 kg less? **Answer:** 70 kg.
Comparison	The fact to which the question refers requires qualitative or quantitative comparison based on an understanding of event timing and/or the time of questioning.	[11.09.2025] **User:** Today I leg pressed 145 kg, last week it was 148 and 150 kg. [30.10.2025] **Question:** What was my best result that month? **Answer:** 150 kg.
UserQA	The fact to which the question refers requires an understanding not only of dialogue-based information but also of general or domain-specific knowledge, in the context of event timing and/or the time of questioning.	[15.03.2025] **User:** It's my birthday today. [...] [15.07.2025] **User:** My husband's 40th birthday is tomorrow. [25.10.2025] **Question:** What zodiac signs are me and my husband? **Answer:** Pisces and Cancer.
Causal Reasoning	The fact to which the question refers requires an understanding of causal relationships in the context of event timing and/or the time of questioning. Condition: To answer a question about the user, it is necessary to synthesize information from multiple sessions.	[15.10.2025] **User:** I argued with my mom, I'm so mad, how do I leave home? [...] [22.10.2025] **User**: We argued again about homework, same as last week. [22.12.2025] **Question:** Why did we argue on October 15? **Answer:** Because of homework.
MultiStep	The fact to which the question refers requires multi-step reasoning in the context of event timing and/or the time of questioning.	[01.09.2025] **User:** I got back from vacation, I was there from the 15th. [...] [15.11.2025] **User:** I'll have a short vacation next month (5 days), and a small one this month from the 1st to the 4th. [30.11.2025] **Question:** How long was my second vacation? **Answer:** 4 days.
ComplexRelations		
EventEvent Time Understanding	The fact to which the question refers is revealed only through another event and requires an understanding of event timing and/or the time of questioning. At least two interrelated events are required.	[15.11.2025] **User:** Vasya and Petya came over. [...] [17.11.2025] **User:** My birthday was the day before yesterday. [30.11.2025] **Question:** Who came over on my birthday? **Answer:** Vasya and Petya.
During Reasoning	A question concerning the overlap of events. At least two events are required.	[20.10.2025] **User:** I started painting the walls. [...] [21.10.2025] **User:** I went for a walk with my nephew Vanya. [...] [22.10.2025] **User:** I'm hanging out with Vanya. [...] [25.10.2025] **User:** I finished painting the walls. [...] [30.10.2025] **User:** I'm with my nephew again, everything's good. [20.11.2025] **Question:** How many times did I see my nephew while painting the walls? **Answer:** 2 times.

Continuation

Category&Type	Type Description	Example
Trends	The fact to which the question refers changes over the course of the dialogue. The question is posed to determine how a particular event or fact evolved over time.	[20.01.2025] **User:** I smoke 15 cigarettes a day. [...] [21.02.2025] **User:** I smoke 10 cigarettes a day. [...] [22.03.2025] **User:** I smoke 5 cigarettes a day. [...] [25.04.2025] **User:** I smoke 1 cigarette a day. [...] [30.05.2025] **User:** I smoke 15 cigarettes a day. [21.09.2025] **Question:** How did my daily smoking change? **Answer:** First it decreased from 15 cigarettes to 1, then increased sharply back to 15 cigarettes.
OTHER TYPES		
Abstention	The dialogue does not contain an answer to this question. This is a control (trick) question intended to prevent hallucinations. The answer to the question is: No such information and similar.	**User:** I'm 17 and I have a boyfriend. **Question:** How old is my boyfriend? **Answer:** There's no information about that.